

PC-24 Study Reference

Consolidated verified data — MSN 631 (MSN 501-UP column throughout).

198 cards from the FSI Memory Flash Cards, the Pilatus AFM (Report 02371, Issue 003 Rev 08) and the ACE Avionics Pilot's Guide, plus 117 independently verified lines from a student gouge.

Every value in this document was checked against its source. Items that could not be verified are excluded.

Emergency memory items

Standard: **verbatim, in order, out loud**. These 16 are the procedures the AFM encloses in a solid red box.

QUESTION	ANSWER
E-1 ALL BRAKES FAIL Both the main brakes and emergency brake system are sensed to be inoperative. QRH 3-BRKS-01	On-ground during taxi: NOTE: Some brake pressure may still be available. 1. Wheel brakes and EMER Brake — Use if available to stop aircraft
E-2 CABIN ALTITUDE Cabin altitude above warning threshold and voice callout "CABIN ALTITUDE". QRH 3-ECS-01	1. Crew oxygen masks — Don/100%/Establish COMM 2. Passenger oxygen — ON, Instruct passengers to don dropdown masks IF rapid or explosive decompression: 3. Emergency Descent-3-NAE-02 — Accomplish
E-3 CABIN PRESSURE Excessive positive or negative differential pressure. QRH 3-ECS-02	In Flight with excessive negative differential pressure: 1. Aircraft — Reduce descent rate
E-4 L ENG FAIL / R ENG FAIL The indicated N1 has dropped 15% within 0.5 seconds. This will activate ATR during takeoff. QRH 3-ENG-02	1. Engine instruments — Monitor / Confirm
E-5 L ENGINE FIRE / R ENGINE FIRE The indicated Engine Fire detector senses a temperature above 510 °C. Additional indications: Audio "Left Engine Fire" callout, RED "FIRE" light on the associated Engine Isolation Pushbutton, Synoptic page: FIRE flag on Engine ITT indication, Red color of associated engine outline. QRH 3-FIRE-01	1. ENGINE FIRE — CONFIRM L or R IF Fire Confirmed: 2. THRUST LEVER — IDLE 3. ENGINE SWITCH — OFF / STOP 4. ENGINE ISOLATION push button — PUSH Check associated CAS "ENGINE ISOLATED" after five sec. CONFIRM ASSOCIATED green "1" illuminated. NOTE: If possible, keep pitch and bank angles to within ±10° to achieve maximum extinguisher effectiveness. IF CAS Message remains, affected side: 5. EXTINGUISHER push button — PUSH
E-6 ENGINE HUNG START ON GROUND Indications: Engine Fails to Accelerate for > 30 Seconds. QRH 3A-NAA-08	1. Engine Switch — OFF
E-7 ENGINE HOT START ON GROUND Indications for Hot Start: Rapidly rising ITT approaching engine limits of 1,000 °C or ITT hanging between 900 °C and 1,000 °C for 15 seconds. QRH 3A-NAA-07	1. Engine Switch — OFF
E-8 SWPS INADVERTENT PUSHER OPERATION Non-commanded pusher operation, rapid nose-down motion. QRH 3A-NAA-18	WARNING: NATURAL STALLS ARE NOT PREVENTED WITH THE STICK PUSHER INOPERATIVE. STALLS MUST BE AVOIDED WHEN THE STICK PUSHER IS INOPERATIVE. 1. Control wheel — Hold against pusher action 2. AP disconnect button — Press and hold continuously
E-9 ENGINE FIRE ON GROUND / TAIL PIPE FIRE If during start, engine fire detector senses a temperature above 510 °C or else engine fire is indicated by ground personnel. QRH 3-NAE-05	1. THRUST LEVER — IDENTIFY/IDLE Affected Engine 2. ENG SWITCH — OFF 3. ENG ISOL pushbutton — Push/check after 5 sec amber ENG ISOL; Confirm green "1" 4. FIRE XTING pushbutton — Push to activate

QUESTION	ANSWER
E-10 ENGINE FIRE OR FAILURE DURING TAKEOFF ENGINE FIRE OR FAILURE DURING TAKEOFF QRH 3-NAE-06	Airspeed below V1: 1. Rejected Takeoff - 3-NAE-19 — Accomplish -END- Airspeed at or above V1: 1. THRUST levers — Fully forward, ATR 2. VR — Rotate to 9° ANU Initially 3. Airspeed — Maintain at or above V2 4. L/G — When positive rate of climb – UP
E-11 REJECTED TAKEOFF REJECTED TAKEOFF QRH 3-NAE-19	CAUTION: IF AUTOTHROTTLE WAS ENGAGED FOR TAKEOFF, ENSURE AUTOTHROTTLE IS DISCONNECTED BY EITHER PRESSING AN AT DISC PUSHBUTTON OR HOLDING THE THROTTLES AT IDLE FOR 2 TO 3 SECONDS UNTIL AUTOTHROTTLE DISCONNECTS. 1. THRUST levers — IDLE 2. BRAKES — Max Braking / As Required
E-12 EMERGENCY DESCENT EMERGENCY DESCENT QRH 3-NAE-02	1. Crew Oxygen Masks — DON / 100% / Establish COMM 2. PASSENGER OXYGEN — ON, instruct passengers to don dropdown masks 3. SEAT BELTS — ON 4. THRUST Levers — IDLE 5. AIRBRAKE lever — OUT
E-13 SMOKE, FIRE, OR FUMES IN COCKPIT / CABIN SMOKE, FIRE, OR FUMES IN COCKPIT / CABIN QRH 3-SMOKE-01	IMMEDIATE ACTION IF SMOKE AND / OR FUMES ARE PERCEPTIBLE: 1. CREW OXYGEN MASK — DON / select EMGCY / Vent valve open 2. Smoke Goggles — DON 3. Passenger OXYGEN — ON, Instruct PAX to don dropdown masks LAND ASAP / INITIATE DIVERSION 4. Aircraft — Land as soon as possible
E-14 EMERGENCY EVACUATION EMERGENCY EVACUATION QRH 3-EVAC-01	1. PARK BRAKE — Set 2. ATC / distress message — Notify 3. ENG L+R switches — Both OFF
E-15 TCAS RESOLUTION ADVISORY Accompanied by audio commands for climb, descend, level off or maintain vertical speed. QRH 3B-FAS-20	1. Autopilot — Use yoke quick disconnect if AP engaged 2. Aircraft attitude — Immediately comply with RA commands indicated on the PFD 3. ATC — Inform of TCAS RA When clear of conflict: 4. Aircraft — Return to previously assigned altitude and advise ATC
E-20 AFCS ABNORMAL DISCONNECT AFCS ABNORMAL DISCONNECT QRH 3A-NAA-01	Indication: Flashing red “AP” on PFD and continuous “Cavalry Charge” aural warning. 1. Airplane control wheel — Grasp firmly and regain aircraft control 2. Autopilot Disengage switch — Press to cancel aural warning (pilot or copilot wheel) 3. Aircraft — Retrim manually as necessary If no AFCS associated CAS messages: 4. Aircraft — Attempt to re-engage autopilot once

Normal procedure flows

Standard: fluent — sequence and every confirmation correct. These share the E- numbering but are **not** memory items.

QUESTION	ANSWER
E-16 RH ENGINE START RH ENGINE START AFM 4-PF-06	RH Engine — Start To start the RH Engine: 1. R Engine switch rotate to RUN 2. Confirm R Booster pump running and no FUEL PRESSURE CAS messages appear 3. R START Press

QUESTION	ANSWER
E-17 RH ENGINE START (CONTINUED) RH ENGINE START (CONTINUED) AFM 4-PF-06	1. Monitor start sequence as follows: 2. When pressing start button "START" will appear "ON" on the PFD 3. At about 10% N2, "START" and "IGN" is illuminated in inverse video. Check N1 is increasing 4. Observe light-up and engine parameters until "IDL" is reached 5. Depending on ambient conditions "START" and "IGN" extinguish at approximately 45% - 50% of N2
E-18 LH ENGINE START LH ENGINE START AFM 4-PF-06	1. LH Engine — Start • Start LH Engine Accordingly NOTE: The right engine must exit QPM and set to Ground Idle before Left Engine start
E-19 ENGINE APPROXIMATE GND IDLE VALUES ENGINE APPROXIMATE GND IDLE VALUES AFM 4-PF-06	N1: 25% ITT: 400 °C N2: Min 53.4% FF: 160-180 lb Confirm GEN 2 takes over and is supplying the aircraft. NOTE: If TAT or reported OAT is <10 °C and visible signs of moisture are present then set the IPS rotary switch to AUTO/NAI.
E-21 CLIMB CLIMB AFM 4-IF-01	1. Gear — UP • Confirm Lever Up and Lights Out on System Summary Page 2. Yaw Damper — ON • Confirm on FMA 3. Flaps — UP • Confirm lever at 0 and indication on System Summary Page 4. Thrust — MCT • Thrust Levers back to MCT gate within five min after setting T/O • Verify MCT indications on both engine PFD indications

Limitations

Standard: **exact number, exact unit.** "About" is a bust.

QUESTION	ANSWER
L-1 MAXIMUM OPERATING SPEED 1. VMO ($\leq$ 28,230 ft) — ? 2. MMO ($\geq$ 28,230 ft) — ? AFM Page 2-1-9	1. VMO — 290 KIAS 2. MMO — 0.74 M NOTE MMO schedule for maximum speed above 28,230 ft at this location.
L-2 WEIGHT LIMITS 1. Max Zero Fuel Weight (MSN 101-500) — ? 2. Max Ramp Weight (MSN 101-500) — ? 3. Max Zero Fuel Weight (MSN 501-UP) — ? 4. Max Ramp Weight (MSN 501-UP) — ? AFM Page 2-1-23	1. Max Zero Fuel Weight (MSN 101-500) — 14,220 lbs (6,450 kg) 2. Max Ramp Weight (MSN 101-500) — 18,400 lbs (8,345 kg) 3. Max Zero Fuel Weight (MSN 501-UP) — 14,660 lbs (6650 kg) 4. Max Ramp Weight (MSN 501-UP) — 18,840 lbs (8545 kg)
L-3 WEIGHT LIMITS 1. Max Takeoff Weight (MSN 101-500) — ? 2. Max Landing Weight (MSN 101-500) — ? 3. Max Takeoff Weight (MSN 501-UP) — ? 4. Max Landing Weight (MSN 501-UP) — ? AFM Page 2-1-23	1. Max Takeoff Weight (MSN 101-500) — 18,300 lbs (8,300 kg) 2. Max Landing Weight (MSN 101-500) — 16,900 lbs (7,665 kg) 3. Max Takeoff Weight (MSN 501-UP) — 18,740 lbs (8,500 kg) 4. Max Landing Weight (MSN 501-UP) — 17,340 lbs (7,865 kg)
L-4 WEIGHT LIMITS 1. Maximum Baggage Weight (Small Restraint Net) — ? 2. Maximum Baggage Weight (Large Restraint Net) — ? 3. Maximum Cargo Weight (MSN 101-500) — ? 4. Maximum Cargo Weight (MSN 500-UP) — ? AFM Page 2-1-23	1. Maximum Baggage Weight (Small Restraint Net) — 400 lbs (180 kg) 2. Maximum Baggage Weight (Large Restraint Net) — 530 lbs (240 kg) 3. Maximum Cargo Weight (MSN 101-500) — 2,500 lbs (1,134 kg) 4. Maximum Cargo Weight (MSN 501-UP) — 2,940 lbs (1,334 kg)

QUESTION	ANSWER
L-5 LOADING LIMITS 1. Maximum Floor Loading (Over Floor Panels) — ? 2. Maximum Floor Loading (On Seat Rails) — ? AFM Page 2-1-23	1. Maximum Floor Loading (Over Floor Panels) — 100 lbs/ft ² (500 kg/m ²) 2. Maximum Floor Loading (On Seat Rails) — 670 lbs/ft (1,000 kg/m)
L-6 LOAD FACTOR LIMITS 1. Flaps Up — ? 2. Flaps Down — ? AFM Page 2-1-24	1. Flaps Up — +3.0g ; -1.2g 2. Flaps Down — +2.0g ; -0.0g
L-7 OPERATING ALTITUDE LIMITS Maximum Operating Altitude — ? AFM Page 2-1-26	Maximum Operating Altitude — 45,000 ft (13,716 m)
L-8 OUTSIDE AIR TEMPERATURE LIMITS 1. Minimum OAT (Sea Level) — ? 2. Maximum OAT (Sea Level) — ? AFM Page 2-1-26	1. Minimum OAT (Sea Level) — -54 °C (-67 °F) 2. Maximum OAT (Sea Level) — +50 °C (122 °F)
L-9 PASSENGER SEATING LIMITS The maximum number of passengers in the Executive Interior Cabin is _____ passengers, including optional fit for two additional infants at first seating row left and right sides. AFM Page 2-1-25	The maximum number of passengers in the Executive Interior Cabin is 8 passengers, including optional fit for two additional infants at first seating row left and right sides.
L-10 AIRPORT LIMITS 1. Maximum Runway Slope — ? 2. Approved Takeoff/Landing Surfaces — ? 3. Contaminated Paved Runways — ? With Gravel Kit Installed: Dry/Wet Dirt-Sand-Gravel — ? Dry/Wet Grass — ? AFM Page 2-1-1, 2-1-2	1. Maximum Runway Slope — ±2% 2. Approved Takeoff/Landing Surfaces — Dry and Wet Paved Runways 3. Contaminated Paved Runways — *Per AFM Supplement 02442 With Gravel Kit Installed: • Dry/Wet Dirt-Sand-Gravel — Per AFM Supplement 02444 • Dry/Wet Grass — Per AFM Supplement 02473
L-11 AIRPORT LIMITS 1. Maximum Tailwind Component for takeoff and landing — ? 2. Minimum Pressure Altitude — ? 3. Maximum Pressure Altitude Landing Field Elevation — ? AFM Page 2-1-2	Maximum Tailwind Component for Takeoff and Landing — 10 kt Minimum Pressure Altitude — -1,000 ft MSL Maximum Pressure Altitude Landing Field Elevation — 10,000 ft MSL
L-12 ENGINES 1. Number of Engines — ? 2. Engine Manufacturer / Model Number — ? AFM Page 1-1-2	1. Number of Engines — 2 2. Engine Manufacturer / Model Number — Williams International FJ44-4A QPM
L-13 OIL QUANTITY Oil level must not be below the _____ indicator mark on the oil tank sight glass prior to flight. AFM Page 2-1-26	Oil level must not be below the ADD indicator mark on the oil tank sight glass prior to flight.
L-14 ENGINE THRUST SETTINGS (LB) 1. Normal Takeoff Thrust — ? 2. Automatic Thrust Reserve — ? AFM Page 1-1-2	1. Normal Takeoff Thrust — 3,420 lbs 2. Automatic Thrust Reserve — 3,600 lbs
L-15 STARTER DUTY CYCLE LIMITS (SECONDS) 1. Start Attempt 1 — ? 2. Dry Motor Run — ? 3. Start Attempt 2 — ? AFM Page 2-1-31, 2-1-32	1. Start Attempt 1 — ON (Start) - 50 Sec., OFF - 1 Min. 2. Dry Motor Run — ON (Start) - 50 Sec., OFF - 1 Min. 3. Start Attempt 2 — ON (Start) - 50 Sec., OFF - 30 Min. NOTE Abort engine start where N1 fails to increase by 25% N2.

QUESTION	ANSWER
L-16 ENGINE STARTING LIMITS 1. Maximum Tailwind Component for Engine Start — ? 2. Maximum Crosswind Component for Engine Start — ? AFM Page 2-1-31	1. Max. Tailwind Component for Engine Start — 20 kt 2. Max. Crosswind Component for Engine Start — 25 kt Tower Wind
L-17 ENGINE STARTING LIMITS 1. Minimum Time Between Engine Starts — ? 2. Maximum Time to Light-off for Engine Start — ? AFM Page 2-1-31	1. Minimum Time Between Engine Starts — 60 Seconds 2. Maximum Time to Light-off for Engine Start — 12 Seconds
L-18 ENGINE STARTING LIMITS 1. Minimum Ni-Cad Battery Voltage for Engine Start — ? 2. Minimum Li-Ion Battery Voltage for Engine Start — ? 3. Minimum Oil Temperature for Engine Start — ? AFM Page 2-1-16	1. Min. Ni-Cad Battery Voltage for Engine Start — 22.0 VDC (BAT 1); 23.5 VDC (BAT 2) 2. Min. Li-Ion Battery Voltage for Engine Start — State of Charge (SOC) indicators of both batteries green 3. Min. Oil Temperature for Engine Start — -40 °C
L-19 POST START OIL TEMPERATURE LIMITATIONS The minimum oil temp for N2 > ___ % or QPM operation is ___ °C. AFM Page 2-1-32	The minimum oil temp for N2 > 80% or QPM operation is +10 °C.
L-20 QPM OPERATION — ENGINE LIMITS Activation of QPM is permitted after at least two minutes of stabilized operation at _____ N2. After QPM termination, run engine at ground idle thrust for at least _____ before shutdown or increasing thrust. AFM Page 2-1-36	Activation of QPM is permitted after at least two minutes of stabilized operation at Ground Idle N2. After QPM termination, run engine at ground idle thrust for at least 2 Minutes before shutdown or increasing thrust.
L-21 QPM OPERATION — ENGINE LIMITS QPM operation is permitted provided the aircraft remains stationary with the _____ on. The thrust lever remains _____ during QPM operation. The maximum generator load while in QPM of _____ is not exceeded. AFM Page 2-1-36	QPM operation is permitted provided the aircraft remains stationary with the Park Brake on. The thrust lever remains IDLE during QPM operation. The maximum generator load while in QPM of 250 A is not exceeded.
L-22 ENGINE COOLDOWN Engine shut-down is permitted when the engine has been run at idle thrust for more than _____. AFM Page 2-1-36	Engine shut-down is permitted when the engine has been run at idle thrust for more than 2 minutes. NOTE It is recommended that the engine bleed valve is closed during this time.
L-23 FUEL QUANTITY 1. Total quantity — ? 2. Usable quantity — ? 3. Maximum permissible imbalance — ? AFM Page 2-1-18	1. Total quantity — 895 US gal/5,999.8 lbs (3,389 liter/2,721 kg) 2. Usable quantity — 890 US gal/5,964 lbs (3,369 liter/2,705 kg) 3. Maximum permissible imbalance — 49 US gal/330 lbs (189 liter/150 kg)
L-24 FUEL TEMPERATURE 1. Maximum fuel temperature — ? 2. Minimum fuel temperature — ? AFM Page 2-1-17	1. Maximum fuel temperature — 80 °C 2. Minimum fuel temperature — -40 °C
L-25 PRESSURE REFUELING Maximum refueling pressure — ? AFM Page 2-1-19	Maximum refueling pressure — 60 PSI
L-26 FUEL MISCELLANEOUS The electric fuel booster pumps and fuel crossfeed valve must be _____. AFM Page 2-1-17	The electric fuel booster pumps and fuel crossfeed valve must be operative.
L-27 FUEL GRADES Approved Fuel Grades — ? AFM Page 2-1-17	Approved Fuel Grades — Jet A, Jet A-1, JP-8, TS-1

QUESTION	ANSWER
L-28 MAXIMUM LANDING GEAR SPEED 1. VLO (Extend) — ? 2. VLO (Retract) — ? 3. VLO (Extend EGES) — ? 4. VLE — ? AFM Page 2-1-10	1. VLO (Extend) — The lower of 250 KIAS or 0.74 M 2. VLO (Retract) — 200 KIAS 3. VLO (Extend EGES) — 180* KIAS 4. VLE — The lower of 250 KIAS or 0.74 M NOTE *QRH L/G Emergency Extension (3-NAE-13) requires airspeed 160 KIAS or less.
L-29 TIRE LIMITS Maximum Tire Speed — ? AFM Page 2-1-15	Maximum Tire Speed — 165 kt groundspeed
L-30 MAXIMUM FLAP EXTENDED SPEED VFE 1. 8° — ? 2. 15° — ? 3. 33° — ? AFM Page 2-1-9	1. 8° — 200 KIAS 2. 15° — 200 KIAS 3. 33° — 175 KIAS
L-31 CABIN PRESSURIZATION LIMITS 1. Maximum Positive Cabin Pressure Differential — ? 2. Maximum Negative Cabin Pressure Differential — ? 3. Maximum Pressure Differential for Takeoff & Landing — ? AFM Page 2-1-15	1. Maximum Positive Cabin Pressure Differential — 9.27 psid 2. Maximum Negative Cabin Pressure Differential — -0.3 psid 3. Maximum Pressure Differential for Takeoff & Landing — 0.7 psid
L-32 AIRSTAIR LIMIT Persons allowed to be on the forward airstair (forward cabin door) at any time — ? AFM Page 2-1-16	Persons allowed to be on the forward airstair (forward cabin door) at any time — 1 Person
L-33 EMERGENCY EXIT LIMIT Each internal emergency exit overnight security pin, if installed, must be ____ and ____ before flight. AFM Page 2-1-16	Each internal emergency exit overnight security pin, if installed, must be removed and stowed before flight.
L-34 GPU LIMITS Prior to connecting a GPU, ensure GPU voltage is greater than ____ voltage. AFM Page 2-1-17	Prior to connecting a GPU, ensure GPU voltage is greater than battery voltage.
L-35 GPU LIMITS 1. Voltage — ? 2. Load Capacity (Initial Start Surge) — ? 3. Load Capacity (Continuous) — ? AFM Page 2-1-16	1. Voltage — 25.0 - 29.5 VDC 2. Load Capacity (Initial Start Surge) — 1,200 A 3. Load Capacity (Continuous) — 450 A
L-36 GPU LIMITS 1. Minimum Volts for Start — ? 2. Minimum Volts to Charge Batteries — ? 3. Maximum Volts to Charge Batteries — ? 4. Minimum OAT for GPU Start — ? AFM Page 2-1-16, 2-1-17	1. Minimum Volts for Start — 24.0 VDC 2. Minimum Volts to Charge Batteries — 28.0 VDC 3. Maximum Volts to Charge Batteries — 29.5 VDC 4. Minimum OAT for GPU Start — -54 °C
L-37 GENERATOR LIMITS 1. QPM — ? 2. Ground Idle and Above — ? 3. Flight — ? 4. Maximum Voltage — ? AFM Page 2-1-16	1. QPM — 250 A 2. Ground Idle and Above — 400 A 3. Flight — 400 A 4. Maximum Voltage — 29.5 VDC

QUESTION	ANSWER
L-38 BATTERY LIMITS 1. Minimum Battery Temperature for Flight — ? 2. Minimum Ni-Cd Battery Temperature — ? 3. Minimum Li-Ion Battery Temperature — ? 4. Minimum Ni-Cd Battery Volts for Start — ? 5. Minimum Li-Ion Battery Volts for Start — ? 6. Maximum Battery Charge Rate Before Takeoff — ? AFM Page 2-1-16	1. Minimum Battery Temperature for Flight — 0 °C 2. Minimum Ni-Cd Battery Temperature — -20 °C 3. Minimum Li-Ion Battery Temperature — -5 °C 4. Minimum Ni-Cd Battery Volts for Start — 22.0 VDC (Battery 1); 23.5 VDC (Battery 2) 5. Min Li-Ion Battery Volts for Start — SOC Indicator on Both Batteries Green 6. Maximum Battery Charge Rate Before Takeoff — Battery 1+2 must be <50 A and decreasing
L-39 AUTOPILOT LIMITS During autopilot operation, a pilot must be ____ in a pilot position with their ____ fastened. AFM Page 2-1-3	During autopilot operation, a pilot must be seated in a pilot position with their seat belt fastened.
L-40 AUTOPILOT LIMITS The autopilot (AP) and yaw damper (YD) must be ____ during takeoff and landing. AFM Page 2-1-3	The autopilot (AP) and yaw damper (YD) must be OFF during takeoff and landing.
L-41 MINIMUM A/P, Y/D, & A/T ALTITUDES 1. All operations except approach, A/P disengage by — ? 2. Non-Precision, Circling, and Visual Approaches (a/s <150 KIAS & VS <1,500 ft/min) — ? 3. Approaches in VGP mode — ? 4. Coupled ILS approaches with G/S < 4.5° — ? AFM Page 2-1-3	1. All operations except approach, A/P disengage by — 1,000 ft AGL 2. Non-Precision, Circling, and Visual Approaches (a/s <150 KIAS & VS <1,500 ft/min) — 400 ft AGL 3. Approaches in VGP mode — 200 ft AGL 4. Coupled ILS approaches with G/S < 4.5° — 200 ft AGL
L-42 YAW DAMPER LIMITS Operation above ____ is prohibited with the Yaw Damper disengaged. AFM Page 2-1-3	Operation above 30,000 feet MSL is prohibited with the Yaw Damper disengaged.
L-43 SEVERE ICING CONDITIONS Flight in freezing rain, freezing drizzle, or mixed icing conditions (supercooled liquid water and ice crystals) may result in ice build-up on protected surfaces exceeding the capability of the ____ system, or may result in ice forming aft of the protected surfaces. AFM Page 2-1-21	Flight in freezing rain, freezing drizzle, or mixed icing conditions (supercooled liquid water and ice crystals) may result in ice build-up on protected surfaces exceeding the capability of the Ice Protection system, or may result in ice forming aft of the protected surfaces.
L-44 AUTOPILOT USAGE IN ICING CONDITIONS Care must be taken when using the autopilot that tactile cues, such as increased aileron forces, could potentially be masked by the autopilot function. Periodically ____ the autopilot to check for abnormal forces. AFM Page 2-1-21	Care must be taken when using the autopilot that tactile cues, such as increased aileron forces, could potentially be masked by the autopilot function. Periodically disengage the autopilot to check for abnormal forces.
L-45 NACELLE ANTI-ICE The Nacelle Anti-ice (NAI) must be on for all ground operations, takeoff, and landing when: AFM Page 2-1-37	The Nacelle Anti-ice (NAI) must be on for all ground operations, takeoff, and landing when: 1. Visible moisture (i.e. standing water, slush, snow on taxi ways / runway) and indicated SAT or reported OAT <10 °C or 2. Ice or other frozen precipitation is observed adhering to any part of the engine inlet.
L-46 NACELLE ANTI-ICE The NAI must be on for climb, cruise, and descent when: AFM Page 2-1-37	The NAI must be on for climb, cruise, and descent when: 1. In visible moisture with TAT <10 °C, or 2. Flight through ice crystals, or 3. Anytime ice accumulation is observed on the airframe
L-47 NACELLE ANTI-ICE Do not operate the NAI when the TAT is ____ °C. Do not operate the NAI during ____ operation. When required for takeoff, NAI must be selected ON prior to setting ____. AFM Page 2-1-37	Do not operate the NAI when the TAT is >10 °C. Do not operate the NAI during QPM operation. When required for takeoff, NAI must be selected ON prior to setting takeoff power.

QUESTION	ANSWER
L-48 HORIZONTAL STABILIZER DE-ICE (HSDI) Operation of the HSDI is not recommended when TAT < ____ unless required due to the presence of icing conditions. If operated at TAT ≤ -40 °C, a ____ must be performed before takeoff. AFM Page 2-1-19	Operation of the HSDI is not recommended when TAT < -40 °C unless required due to the presence of icing conditions. If operated at TAT ≤ -40 °C, a visual inspection of the pneumatic boots and a system functional test must be performed before takeoff.
L-49 OXYGEN SYSTEM LIMITS 1. Minimum oxygen quantity for dispatch — ? 2. Minimum oxygen supply for each occupant for pressurized flight above 25,000 feet MSL — ? AFM Page 2-1-28	1. Minimum oxygen quantity for dispatch — 680 Liters 2. Minimum oxygen supply for each occupant for pressurized flight above 25,000 feet MSL — 10 minutes
L-50 ELECTRONIC CHARTS Pilots are responsible for taxiing by ____ and for flying by airborne navigation with use of ____. AFM Page 2-1-7	Pilots are responsible for taxiing by external visual references and for flying by airborne navigation with use of primary navigation instruments.
L-51 ELECTRONIC CHARTS Position accuracy of a chart aircraft symbol can decrease in the case of insufficient ____ or _____. The aircraft symbol is not in-scale with ACE™ Electronic Charts. AFM Page 2-1-7	Position accuracy of a chart aircraft symbol can decrease in the case of insufficient GPS signal reception or GPS sensor failure. The aircraft symbol is not in-scale with ACE™ Electronic Charts.
L-52 ELECTRONIC CHARTS ACE™ Electronic Charts do not replace approved ____ or approved _____. AFM Page 2-1-7	ACE™ Electronic Charts do not replace approved published papers or approved electronic systems for aeronautical charts.
L-53 WEATHER RADAR Operation of the weather radar is prohibited: AFM Page 2-1-8	Operation of the weather radar is prohibited: 1. Within a hangar or other enclosure 2. During airplane refueling/defueling operations 3. With personnel present within the acceptable safe distance and ± 60° from the aircraft nose
L-54 INAV MAP 1. The INAV topographical map shall not be used for primary ____. 2. The display of airspace shall not be used as the ____. AFM Page 2-1-8	1. The INAV topographical map shall not be used for primary navigation. 2. The display of airspace shall not be used as the sole airspace reference.
L-55 SMART VIEW (SV) Operations using Smart View are permitted when it is not used as a ____ of navigation reference or a ____ reference for obstacle clearance. AFM Page 2-1-8	Operations using Smart View are permitted when it is not used as a primary source of navigation reference or a primary reference for obstacle clearance.
L-56 VERTICAL SITUATION DISPLAY Use of the vertical situation display is permitted provided it is not used as a ____ of navigation. AFM Page 2-1-8	Use of the vertical situation display is permitted provided it is not used as a primary source of navigation.
L-57 FMS Accomplishment of ____ approach types using the FMS is prohibited. AFM Page 2-1-4	Accomplishment of ILS, LOC, LOC-BC, LDA, SDF and MLS approach types using the FMS is prohibited.

Aircraft general

Standard: fluent.

QUESTION	ANSWER
AGI-1 DISPLAY UNIT (DU) COLOR CONVENTIONS — RED DISPLAY UNIT (DU) COLOR CONVENTIONS RED — ? Honeywell Pilot's Guide	Warning Conditions, Warning CAS Messages; Flight Envelope and System Limitations, Invalid Primary Instruments
AGI-2 DISPLAY UNIT (DU) COLOR CONVENTIONS — AMBER DISPLAY UNIT (DU) COLOR CONVENTIONS AMBER — ? Honeywell Pilot's Guide	Caution Conditions, Caution CAS Messages; Invalid Data, Miscompare Annunciation
AGI-3 DISPLAY UNIT (DU) COLOR CONVENTIONS — MAGENTA DISPLAY UNIT (DU) COLOR CONVENTIONS MAGENTA — ? Honeywell Pilot's Guide	FD Coupled Data, Active Route/Flight Plan, Flight Director Cue
AGI-4 DISPLAY UNIT (DU) COLOR CONVENTIONS — CYAN DISPLAY UNIT (DU) COLOR CONVENTIONS CYAN — ? Honeywell Pilot's Guide	Advisory CAS Messages; Crew selected Data; Window with focus or cursor, curled arrow Prompts; FMS pending Flight Plan
AGI-5 DISPLAY UNIT (DU) COLOR CONVENTIONS — GREEN DISPLAY UNIT (DU) COLOR CONVENTIONS GREEN — ? Honeywell Pilot's Guide	Autopilot Engaged Modes; Current Data; ON or Functioning
AGI-6 DISPLAY UNIT (DU) COLOR CONVENTIONS — WHITE DISPLAY UNIT (DU) COLOR CONVENTIONS WHITE — ? Honeywell Pilot's Guide	Armed/Standby Modes; Status Data; Scales and Associated Figures, Labels; Informational CAS Messages; Joystick Cross-Pointer Cursor, Active Flight Plan Waypoints
AGI-7 DISPLAY UNIT (DU) COLOR CONVENTIONS — GRAY DISPLAY UNIT (DU) COLOR CONVENTIONS GRAY — ? Honeywell Pilot's Guide	Legends, Symbols or Units, Boxes (limits for displaying information); Aircraft Structures
AGI-8 DIMENSIONS DIMENSIONS 1. Total Length (Nose to Tail) — ? 2. Total Height (At Tail) — ? 3. Wingspan — ? AFM Page 1-1-3	1. Total Length (Nose to Tail) — 55 ft. 2 in. (16.8 m) 2. Total Height (At Tail) — 17 ft. 4 in. (5.3 m) 3. Wingspan — 55 ft. 9 in. (17 m)
AGI-9 ENGINE DANGER AREAS ENGINE DANGER AREAS 1. Radius Forward of Engines — ? 2. Engine Start — ? 3. Engine Max Thrust — ? AFM Page 1-1-5	1. Radius Forward of Engines — 20 ft (6.1 m) 2. Engine Start — 60 ft (18.3 m) x 36.7 ft (11.2 m) aft of engine 3. Engine Max Thrust — 286 ft (87.2 m) x 125.7 ft (38.3 m) aft of engine

AFM extended

Working knowledge. Built from the AFM to close gaps in the issued deck — mostly limitations.

MSN 631

QUESTION	ANSWER
AFM-1 WEIGHT LIMITS — OUR AIRCRAFT (MSN 631) For MSN 631, state all five weight limits: 1. Max Ramp Weight — ? 2. Max Takeoff Weight — ? 3. Max Landing Weight — ? 4. Max Zero Fuel Weight — ? 5. Max Cargo Weight — ? AFM Table 2-1-15 (p 2-1-23)	MSN 631 is in the MSN 501-up column: 1. Max Ramp Weight — 18,840 lb (8,545 kg) 2. Max Takeoff Weight — 18,740 lb (8,500 kg) 3. Max Landing Weight — 17,340 lb (7,865 kg) 4. Max Zero Fuel Weight — 14,660 lb (6,650 kg) 5. Max Cargo Weight — 2,940 lb (1,334 kg) NOTE: These are the higher values. The 101-500 figures are each 440 lb lower (except baggage/floor limits, which are common to all serials).
AFM-2 CG LIMITS — OUR AIRCRAFT (MSN 631) For MSN 631, state the forward and aft CG limits (% MAC) at: 1. 18,840 lb (Max Ramp Weight) — ? 2. 18,740 lb (MTOW) — ? 3. 13,560 lb — aft limit ? 4. 11,510 lb — forward limit ? AFM Table 2-1-16 (p 2-1-24)	1. 18,840 lb (MRW) — fwd 25.0% / aft 39.8% MAC 2. 18,740 lb (MTOW) — fwd 25.0% / aft 40.0% MAC 3. 13,560 lb (6,150 kg) — aft limit 47.6% MAC 4. 11,510 lb (5,221 kg) — fwd limit 37.0% MAC NOTE: Straight-line variation between points. Datum is 146.1 in (3,711 mm) forward of the forward jacking point.

Airspeeds

QUESTION	ANSWER
AFM-3 VO — MAXIMUM OPERATING MANEUVERING SPEED Do not make full or abrupt control movements above VO. State VO for MSN 501-up MTOW, and the schedule down to minimum weight. AFM Table 2-1-5 (p 2-1-9), 19.3 (p 2-1-24)	18,740 lb (8,500 kg) MTOW MSN 501-up — 170 KIAS 18,300 lb (8,300 kg) MTOW MSN 101-500 — 170 KIAS 17,650 lb (8,005 kg) — 165 KIAS 17,000 lb (7,711 kg) — 160 KIAS 16,000 lb (7,258 kg) — 155 KIAS 15,000 lb (6,804 kg) — 150 KIAS 14,000 lb (6,350 kg) — 145 KIAS 13,000 lb (5,897 kg) — 140 KIAS 12,000 lb (5,443 kg) — 135 KIAS 11,440 lb (5,190 kg) — 120 KIAS NOTE: Avoid rapid, large alternating control inputs — especially combined with large pitch/roll/yaw changes — as these may cause structural failure at ANY speed, including below VO.
AFM-4 VO RUDDER — RUDDER MANEUVERING LIMITS 1. Full/abrupt rudder input limit up to 20,000 ft — ? 2. Full/abrupt rudder input limit above 20,000 ft — ? 3. Speed above which pedal travel is limited to 1/3 — ? 4. Speed above which pedal travel is limited to 1/10 — ? AFM Table 2-1-6 (p 2-1-9)	1. Up to 20,000 ft (6,096 m) — 130 KIAS 2. Above 20,000 ft (6,096 m) — 125 KIAS 3. 185 KIAS (SL to 45,000 ft) — do not exceed 1/3 of pedal travel 4. 290 KIAS (SL to 45,000 ft) — do not exceed 1/10 of pedal travel
AFM-5 VMCG — MINIMUM CONTROL SPEED ON GROUND VMCG (flaps 8° / 15°) — ? AFM Table 2-1-10 (p 2-1-10)	VMCG — 87 KIAS (flaps 8° / 15°)
AFM-6 VMC — MINIMUM CONTROL SPEED FOR TAKEOFF State VMC by weight for flaps 15° and flaps 8°. AFM Table 2-1-11 (p 2-1-10)	Flaps 15°: >=13,228 lb (6,000 kg) — 77 KIAS >12,125 lb (5,500 kg) — 81 KIAS >11,023 lb (5,000 kg) — 84 KIAS Flaps 8°: >=13,669 lb (6,200 kg) — 82 KIAS >12,125 lb (5,500 kg) — 87 KIAS >11,023 lb (5,000 kg) — 90 KIAS NOTE: Flight testing determined the PC-24 has NO critical engine for VMC purposes.

QUESTION	ANSWER
AFM-7 VMCL — MINIMUM CONTROL SPEED FOR LANDING State VMCL by weight for flaps 15° and flaps 33°. AFM Table 2-1-12 (p 2-1-10)	Flaps 15°: >=13,007 lb (5,900 kg) — 78 KIAS >12,125 lb (5,500 kg) — 81 KIAS >11,023 lb (5,000 kg) — 84 KIAS Flaps 33°: >=13,228 lb (6,000 kg) — 73 KIAS >12,125 lb (5,500 kg) — 77 KIAS >11,023 lb (5,000 kg) — 80 KIAS NOTE: No critical engine for VMC purposes.
AFM-8 AIRSPEED TAPE MARKINGS (PFD / ESIS) Identify each airspeed tape marking and its value: 1. Red/white barber pole — ? 2. Placard "G" — ? 3. Placard "Gr" — ? 4. Placards "08" / "15" — ? 5. Placard "33" — ? 6. Red low-speed tape — ? AFM Table 2-1-13 (p 2-1-11)	1. Barber pole (across and upward, right side of tape) — the lower of 290 KIAS or 0.74 M (VMO / MMO) 2. "G" — 250 KIAS, VLO Extend (max landing gear operating speed, extend) 3. "Gr" — 200 KIAS, VLO Retract 4. "08" / "15" — 200 KIAS, VFE at flaps 8° / 15° 5. "33" — 175 KIAS, VFE at flaps 33° 6. Red low speed awareness tape — extends up from the bottom of the tape to shaker speed NOTE: The red high-speed strip extends down from VMO/MMO to the valid VLO or VFE; it is not shown in clean configuration or with gear extended only.

AFCS

QUESTION	ANSWER
AFM-9 AFCS — MINIMUM ENGAGEMENT HEIGHTS AFTER TAKEOFF 1. Minimum autopilot engagement height after takeoff — ? 2. Minimum yaw damper engagement height after takeoff — ? 3. What must be done before YD engagement — ? AFM 4.2.2 (p 2-1-3)	1. Autopilot — 400 ft AGL 2. Yaw damper — 50 ft AGL 3. The landing gear must be selected UP prior to YD engagement
AFM-10 AFCS — GENERAL PROHIBITIONS AND OVERRIDES State the AFCS general limitations: override, tactile feedback, quick disconnect, TCS, and the yaw-damper altitude limit. AFM 4.2.1 (p 2-1-3)	- Do NOT override the autopilot or yaw damper - Do NOT engage the autopilot while the tactile feedback system is active - The pilot-side quick disconnect button must be operable before departure - Servos may be temporarily disengaged without disengaging the AP by pushing and holding the TCS switch on the control wheel for the desired duration - Flight above 30,000 ft MSL with the yaw damper disengaged is PROHIBITED - During AP operation a qualified pilot must be seated in a pilot position with seat belt fastened
AFM-11 AUTOTHROTTLE — OVERRIDE AND APPROVED PHASES 1. How is the autothrottle intentionally overridden — ? 2. In which phases of flight may the autothrottle be used — ? AFM 4.2.1 (p 2-1-3)	1. Reposition and HOLD the thrust lever for a minimum of 3 seconds. This causes an AT disconnect and aural warning, cancelled with the AT quick disconnect button on the power control lever. 2. Takeoff, climb, cruise, descent and approach.
AFM-12 TCAS II — AURAL INHIBIT AND OEI 1. RA and TA aural messages are inhibited below what radio altitude during ascent — ? 2. ...and during descent — ? 3. With one engine inoperative, which TCAS mode is selected — ? 4. May the pilot deviate from an ATC clearance to comply with an RA — ? AFM 4.5 (p 2-1-7)	1. Ascent — inhibited below 1,100 ft radio altitude 2. Descent — inhibited below 900 ft radio altitude 3. One engine inoperative — select TA ONLY 4. Yes — the pilot is authorized to deviate from ATC to the extent necessary to comply with an RA. Prompt return to the ATC cleared altitude when "CLEAR OF CONFLICT" is announced. NOTE: Do not maneuver on a TA alone — TA information is only an aid to visual acquisition.

Powerplant

QUESTION	ANSWER
AFM-13 ENGINE OPERATING LIMITS — THRUST, N1, N2, ITT State thrust, N1, N2 and ITT limits with time limits for: 1. Normal Takeoff — ? 2. ATR — ? 3. Max Continuous / Climb — ? AFM Table 2-1-18 (p 2-1-29), 24.1 (p 2-1-28)	1. Normal Takeoff — 3,420 lb · N1 104.7% · N2 100.8% · ITT 855 °C · max 5 min 2. ATR — 3,600 lb · N1 104.7% · N2 100.8% · ITT 855 °C · max 10 min with OEI, otherwise 5 min 3. Max Continuous / Climb — N1 104.7% · N2 100.8% · ITT 835 °C Oil for all three: PRESS 40-120 psi · TEMP 10-135 °C Except in an emergency, selecting thrust greater than NTO is PROHIBITED. If ATR Disarm CAS is shown, do not attempt takeoff if performance was calculated with ATR available.
AFM-14 ENGINE TRANSIENT LIMITS State the transient (short-duration) limits: 1. N1 — ? 2. N2 — ? 3. ITT — ? 4. Oil pressure high / low — ? 5. Oil temperature — ? AFM Table 2-1-18 (p 2-1-29)	1. N1 — 105.7% (2 min) 2. N2 — 101.5% (2 min) 3. ITT — 855 °C (0 sec) 4. Oil pressure — 130 psi high (max 5 min); 23 psi transient low 5. Oil temperature — 149 °C (when operating below 80% N2 for up to 5 min)
AFM-15 OIL PRESSURE AND TEMPERATURE LIMITS 1. Minimum oil pressure at or above 80% N2 — ? 2. Minimum oil pressure below 80% N2 — ? 3. Maximum allowable oil pressure — ? 4. Oil temp range at ground/flight idle — ? 5. QPM oil pressure range — ? AFM Table 2-1-18 (p 2-1-29)	1. At or above 80% N2 — 40 psig minimum 2. Below 80% N2 — 30 psig minimum 3. 130 psig maximum, for 5 minutes maximum 4. Ground or Flight Idle (continuous) — oil temp -40 to 135 °C; oil press 30 min / 120 max 5. QPM — oil press 30-120 psi, oil temp 10-135 °C, N1 45.4% NOTE: Elevated oil pressure is typically observed when oil temperature is cold.
AFM-16 ZERO AND NEGATIVE G LIMIT The engine is limited to how long at zero and/or negative G, and what indication is acceptable during that time? AFM 24.1 (p 2-1-28)	10 seconds of continuous flight at zero and/or negative G. During this time it is acceptable for engine OIL PRESSURE to indicate ZERO. Exceeding the limit may cause engine damage from oil starvation.
AFM-17 OIL QUANTITY AND CONSUMPTION 1. Oil tank total volume — ? 2. Oil tank fill volume — ? 3. Oil tank usable volume — ? 4. Maximum permissible oil consumption rate — ? 5. Minimum level before flight — ? AFM 20.1, 20.2 (p 2-1-25, 2-1-26)	1. Total volume — 5.85 qt 2. Fill volume — 5.65 qt 3. Usable volume — 4.32 qt 4. Max consumption — 0.032 gal/hr (0.128 qt/hr) 5. Oil level must not be below the ADD indicator mark on the sight glass Approved oils: Mobil Jet II (preferred) and Mobil 254, both MIL-L-23699. Mixing approved oils when topping off is permitted.
AFM-18 HIGH TAILWIND — N1 RESTRICTION When the tailwind component is greater than 10 kt, what N1 restriction applies? AFM 24.1 (p 2-1-28), Table 2-1-19 footnote (p 2-1-31)	CAUTION: When the tailwind component is greater than 10 kt, do NOT exceed 60% N1 until the aircraft rolling speed is greater than the tailwind component. NOTE: Higher minimum throttle-up speeds up to 60% N1 may be required if the tailwind component exceeds 10 kt.
AFM-19 DRY MOTOR RUN — DUTY CYCLE EXCEPTION An engine emergency or abnormal procedure calls for a Dry Motor Run. Do the 1-minute and 30-minute starter cooling waits apply? AFM 24.2.2, Table 2-1-19 (p 2-1-31, 2-1-32)	CAUTION: If any engine related emergency or abnormal procedure requires a Dry Motor Run, the Dry Motor Run may be performed WITHOUT DELAY. There is NO requirement to wait 1 minute (after the first start attempt) or 30 minutes (after the second start attempt) before initiating the Dry Motor Run. Abort engine start where N1 fails to increase by 25% N2. Provided the table wait times are met, an indefinite number of start attempts is permitted.
AFM-20 BEFORE LEFT ENGINE START — DOORS What must be confirmed about the doors before left engine start? AFM 24.2.1 (p 2-1-31)	The main passenger door and the cargo door must be CLOSED and SECURED before left engine start.
AFM-21 FREEZING FOG — ENGINE LIMITATION Engine operation in freezing fog is prohibited below what temperature? AFM 24.7.2 (p 2-1-37)	Do NOT operate the engines in freezing fog conditions below -15 °C.

QUESTION	ANSWER
AFM-22 QPM — COMPLETE PREREQUISITE GATE LIST List every condition that must be satisfied before activating Quiet Power Mode. AFM 24.6, Fig. 2-1-8 (p 2-1-36)	ALL of the following must be true: 1. Aircraft stationary with PARK BRAKE ON 2. Ground icing conditions are NOT present 3. NOT (visible moisture present AND TAT less than 10 °C) 4. Crosswind / tailwind less than 20 kt (average) 5. Only the RIGHT engine is to be operated in QPM 6. A flight crew member seated at the controls with lap belt fastened 7. Aircraft at Ground Idle N2, stabilized, for at least 2 min During QPM: thrust lever at IDLE · generator load ≤250 A · NAI must NOT be operated · right engine must exit QPM and be set to Ground Idle before left engine start. After QPM termination: acceleration or shutdown only after ≥2 min at Ground Idle.
AFM-23 ENGINE FIRE PROTECTION — DISPATCH May the aircraft be operated with one engine fire protection system inoperative? AFM 24.4 (p 2-1-36)	No. Do NOT operate the aircraft system if the left or right fire protection system is inoperative.

Icing

QUESTION	ANSWER
AFM-24 MINIMUM AIRSPEED — FLAPS RETRACTED IN ICING The minimum airspeed with flaps retracted in icing conditions — ? AFM 16.3 (p 2-1-21)	Minimum airspeed, flaps retracted, in icing conditions — 150 KIAS
AFM-25 WING ANTI-ICE (WAI) — INHIBIT AND LIMITS 1. WAI is inhibited under what two conditions — ? 2. Above what TAT must WAI not be operated — ? AFM 15.2 (p 2-1-20)	1. WAI is inhibited when: TAT >15 °C, or the aircraft is on the ground. 2. Do NOT operate WAI with a TAT of more than 10 °C, unless required by the presence of icing conditions.
AFM-26 WINDSHIELD EMERGENCY HEAT — LIMITS When must windshield emergency heating not be operated? AFM 15.3 (p 2-1-20)	Do NOT operate windshield EMERGENCY heating on the ground, or when TAT is >10 °C.
AFM-27 FLAPS IN ICING CONDITIONS 1. Maximum flap extension in icing — ? 2. Flap setting for approach and landing in active icing — ? 3. Retraction limit with ice on the lifting surfaces — ? AFM 16.3 (p 2-1-21)	1. Do NOT extend the flaps beyond 15° in icing conditions. 2. Use FLAPS 15 for approach and landing in current icing conditions (ice detectors indicating ICE or pilot assessment). Do NOT use Flaps 33 for landing in current, active icing conditions. 3. Do NOT retract the flaps below 8° if signs of ice are present on the lifting surfaces. Minimum airspeed with flaps retracted in icing — 150 KIAS.
AFM-28 SEVERE ICING — VISUAL CUES AND AUTOPILOT 1. Name the two visual cues that require you to immediately exit icing. 2. What is the autopilot limitation in severe icing — ? AFM 16.4, 16.5 (p 2-1-21)	1. Immediately exit icing conditions if either cue exists: • Unusual, extensive ice accumulation on the airframe or on the cockpit SIDE WINDOWS • Excessive accumulation of ice BEYOND the wing protected leading edge 2. Do NOT engage the autopilot in severe icing conditions. When the AP is used in icing, periodically disengage it to check for abnormal forces — tactile cues such as increased aileron force can be masked by the autopilot.
AFM-29 ICING — WING CHECK, HOLDING, AND SWPS OVERRIDE 1. What must the pilot periodically check while flying in icing — ? 2. What holding restriction applies in icing — ? 3. When must the SWPS ICE OVRD not be used — ? AFM 16, 16.2, 16.6 (p 2-1-20, 2-1-21)	1. WARNING: THE PILOT SHALL PERIODICALLY CHECK THE LEFT WING UPPER SURFACE ON A REGULAR BASIS WHILE FLYING IN ICING CONDITIONS TO ENSURE THERE IS NO UNUSUAL ICE ACCUMULATION. 2. Extended holding in icing conditions on a SINGLE BLEED SOURCE is not allowed. 3. Do NOT operate the SWPS ICE OVRD push button / SWPS ICE MODE rotary selector if there are signs of ice on the wings.

QUESTION	ANSWER
AFM-30 TAKEOFF WITH CONTAMINATION / FREEZING PRECIPITATION 1. What inspection is required before takeoff if contamination is suspected — ? 2. What must you do on encountering freezing rain or freezing drizzle — ? 3. May you operate into an airport reporting freezing rain — ? AFM 16, 16.1 (p 2-1-20)	1. Do NOT take off if there are signs of ice, snow or frost on the lifting surfaces. A VISUAL AND TACTILE inspection of the wing leading edge and upper wing surface must be conducted. 2. Immediately EXIT the freezing rain or freezing drizzle by changing altitude or course. 3. No. Do NOT operate into airports reporting freezing rain or freezing drizzle. Approved fluids: AMS 1424/1 Type I and AMS 1428/1 Types II, III and IV.
AFM-31 NAI — MINIMUM ON TIME Why is NAI recommended to remain on for a minimum period once activated, and how long? AFM 24.7.1 (p 2-1-37)	NOTE: Turning NAI on and off in quick succession in flight may cause a "No Dispatch" annunciation. It is recommended that NAI remains ON for a minimum of 2 MINUTES when activated, to let the Tt2 sensor heat stabilize and clear all moisture from the probe. Anticipate ice-crystal icing by operating NAI near unstable cumuliform (convective) and high-altitude stratiform clouds.
AFM-32 ICE DETECTION AND WING INSPECTION LIGHT 1. Is the PFID system operative on the ground — ? 2. When must the wing inspection light be operational — ? AFM 15.4, 15.5 (p 2-1-20)	1. No. The PFID system is not operative on the ground, as ice detector signals are not available on the ground. Signals become available as soon as the aircraft is in the air. 2. The wing inspection light must be operational if flying into known icing conditions at night, or if known icing conditions are forecast at night.

Flight controls

QUESTION	ANSWER
AFM-33 AIRBRAKE LIMITATIONS 1. May you land with the airbrake extended — ? 2. By what height must the airbrake be stowed on approach — ? AFM 26.1 (p 2-1-39)	1. No. Do NOT attempt to land with the airbrake extended. 2. Use of the airbrake on approach is permitted provided the aircraft is on a STABLE approach and the airbrake is stowed NO LATER THAN 50 ft AGL.
AFM-34 FLAP AND STAB TRIM LIMITATIONS 1. Above what altitude must flaps not be operated in flight — ? 2. Is landing with Flaps 0° or 8° permitted — ? 3. When may secondary stab trim be used — ? AFM 26.2, 26.3 (p 2-1-39)	1. Do NOT operate the flaps in flight above 20,000 ft. 2. Landing with Flaps 0° or 8° is PROHIBITED unless the flaps have failed. 3. Do NOT use secondary stab trim unless the primary stab trim has failed.
AFM-35 MANEUVER LIMITS AND BANK ANGLE 1. Are acrobatic maneuvers or spins authorized — ? 2. To what bank angle is operation limited — ? 3. Load factor limits — ? AFM 19.3, 19.4 (p 2-1-24)	1. No acrobatic maneuvers, including spins, are authorized. 2. Operation is limited to maneuvers incident to normal flying, stalls (except whip stalls) and steep turns with bank angle NOT MORE THAN 60°. 3. Flaps up +3.0 g / -1.2 g · Flaps down +2.0 g / -0.0 g

Brakes & gear

QUESTION	ANSWER
AFM-36 BRAKE COOLING REQUIREMENT After which two events must the aircraft remain on the ground for a cooling period, and for how long? AFM 7.1 (p 2-1-15)	The aircraft must remain on the ground for at least 120 MINUTES following either of these events: • Rejected takeoff with brake-on speed greater than VR - 20 kt AND heavy brake usage • 0° flap full-stop landing AND heavy brake usage
AFM-37 TIRE LIMITS AND TOWING 1. Maximum permissible tire speed — ? 2. What type must the nose tire be — ? 3. Maximum weight of a towbar-less tug — ? AFM 7.2, 18 (p 2-1-15, 2-1-22)	1. 165 kt groundspeed 2. The nose tire must be a "DUAL CHINE" tire 3. 5,379 lb (2,440 kg) NOTE: Tug weight influences the force on the nose landing gear if the towbar-less tug needs to brake.

Fuel

QUESTION	ANSWER
AFM-38 FUEL QUANTITY — FULL TABLE 1. Total quantity — ? 2. Usable quantity — ? 3. Max permissible imbalance — ? 4. Unusable fuel quantity — ? 5. On what density are these based — ? AFM Table 2-1-14 (p 2-1-18)	1. Total — 895 gal (3,389 l) / 5,999.8 lb (2,721 kg) 2. Usable — 890 gal (3,369 l) / 5,964 lb (2,705 kg) 3. Max imbalance — 49 gal (189 l) / 330 lb (150 kg) 4. Unusable — 5.3 gal (20 l) / 35 lb (16 kg) 5. Based on fuel temp 59 °F (15 °C) and density 6.7 lb/gal (0.803 kg/l). The weight values are not limiting — higher fuel weights are permitted provided the aircraft stays within the weight limits.
AFM-39 REFUELING AND DEFUELING LIMITS 1. Maximum refueling pressure — ? 2. Maximum defueling suction pressure — ? 3. What must be done with the refuel power switch before departure — ? AFM 13.8 (p 2-1-18)	1. Maximum refueling pressure — 60 psi 2. Maximum defueling suction pressure — 10 psi 3. The power switch (PWR - REFUEL / DEFUEL) must be OFF and GUARDED prior to departure
AFM-40 APPROVED FUEL GRADES — FULL LIST List every approved fuel grade with its specification. AFM 13.1, 13.2 (p 2-1-17)	Jet A — ASTM D1655 Jet A-1 — ASTM D1655 JP-8 — MIL-DTL-83133 TS-1 — GOST 10227 Chinese No. 3 — GB 6537-2018 For Chinese No. 3, these additives are PROHIBITED on the PC-24: T1502 (antistatic), T1602 (antiwear), Ethylene Glycol Methyl Ether (anti-icing). NOTE: Fuel meeting ASTM D7566 (synthesized hydrocarbons), redesignated as ASTM D1655, is acceptable. The electric fuel booster pumps and the fuel crossfeed valve must be OPERATIVE. Fuel temp limits -40 °C to +80 °C.

Oxygen

QUESTION	ANSWER
AFM-41 OXYGEN SYSTEM — ADDITIONAL LIMITS 1. Minimum quantity for dispatch — ? 2. Below what altitude may the oxygen saver function not be used — ? 3. Maximum cabin altitude for passenger masks — ? 4. PBE cockpit temperature limit — ? 5. Crew mask press-to-test temperature limit after cold soak — ? AFM 23.1 (p 2-1-28)	1. 680 liters (and, above 25,000 ft MSL pressurized, 10 min supply per occupant) 2. Do NOT use the oxygen saver function below 25,000 ft 3. Passenger oxygen masks are limited to a maximum cabin altitude of 40,000 ft 4. Do NOT operate the PBE when cockpit temperature is ≤ -29 °C (≤ -20 °F) 5. After ground cold soak, do NOT operate the crew oxygen mask press-to-test / reset button when cockpit temperature is ≤ -12 °C (≤ 10 °F) Cylinder must be filled with Aviators Oxygen per MIL-PRF-27210.
AFM-42 OXYGEN — HIGH FIELD MODE What oxygen requirement applies when High Field Mode is active? AFM 23.2 (p 2-1-28)	Single pilot operations — the pilot is required to use oxygen CONTINUOUSLY when High Field Mode is active. Multi pilot operations — at least ONE pilot is required to use oxygen continuously when High Field Mode is active.

Cabin & loading

QUESTION	ANSWER
AFM-43 CABIN PRESSURIZATION — WARNING RANGE Between what values does the maximum cabin NEGATIVE pressure differential exceedance warning occur? AFM 8 (p 2-1-15)	The negative differential warning occurs between -0.3 and -0.6 psid. Limits: max positive differential 9.27 psid · max negative -0.3 psid · max differential for takeoff and landing 0.7 psid.
AFM-44 BAGGAGE VS CARGO — DEFINITIONS AND SECURING 1. What mass distinguishes baggage from cargo — ? 2. How must each be secured — ? 3. Minimum tie-down strap strength — ? 4. What may be stowed unstrapped — ? AFM 6, 6.2 (p 2-1-12, 2-1-15)	1. Baggage is any item ≤ 66 lb (30 kg). Cargo is any item > 66 lb (30 kg). 2. Baggage must be stowed in the restraint system (baggage net). Cargo must be tied down to the seat rails and secured individually with an approved cargo tie down, against a retaining bar secured laterally to the seat rails. 3. Tie-down straps with a breaking strength of at least 1,800 lb per strap. 4. Items up to a total weight of 66 lb (30 kg) may be stowed in the cabin unstrapped, provided a cargo net is installed in front of them. No cargo on the seats; cargo must permit free access to the passenger door and the emergency overwing exits.

QUESTION	ANSWER
AFM-45 BAGGAGE / CARGO CLEARANCE AND CARGO DOOR 1. Clear area forward of the LARGE baggage restraint — ? 2. Clear area forward of the SMALL baggage restraint — ? 3. Clear area in front of cabin cargo — ? 4. Crosswind limit for cargo door operation / loading — ? AFM 6, 6.1 (p 2-1-12, 2-1-13)	1. 9.1 in (250 mm) forward of the large baggage restraint system 2. 7.5 in (190 mm) forward of the small baggage restraint system 3. 4 in (100 mm) in front of the cargo in the cabin 4. Do NOT open / close the cargo door or perform cargo loading operations in crosswinds exceeding 60 kt A life raft carried in the baggage area must be properly secured and, when required by operating regulations, accessible in flight.
AFM-46 PASSENGER SEATING BY CONFIGURATION State the maximum number of passengers for: 1. Executive interiors — ? 2. EX-5S-DIV-1 — ? 3. EX-5S-COM-1S-DIV-1 and EX-6S-DIV-1 — ? 4. Commuter interior — ? 5. What restrictions apply to children and divans — ? AFM 19.5 (p 2-1-25)	1. Executive interiors — 8 passengers (one per seat); optional fit allows two additional infants at the first seating row, left and right 2. EX-5S-DIV-1 — 7 passengers (one per seat, two on the divan) 3. EX-5S-COM-1S-DIV-1 and EX-6S-DIV-1 — 8 passengers (one per seat, two on the divan) 4. Commuter interior — 10 passengers (one per seat) 5. Children only allowed in rows 1 and 2. During taxi, takeoff and landing: no occupant in the forward divan seat; no infants or children in the divan seats; no pregnant women in the divan seats (aft-facing executive seats recommended). In single pilot operation an additional passenger may occupy the RH cockpit seat.

Electrical

QUESTION	ANSWER
AFM-47 BATTERY CHARGING VOLTAGE CEILING The voltage for charging the batteries must not exceed what value? AFM 11.2 (p 2-1-16)	The voltage for charging the batteries must not exceed 32 V. Related: generator max voltage 29.5 Vdc · GPU 25.0-29.5 Vdc · min voltage to charge batteries 28.0 Vdc · max voltage to charge batteries 29.5 Vdc.
AFM-48 CIRCUIT BREAKERS — RESET RULES 1. May CBs be reset in flight — ? 2. How many reset attempts on an Essential Bus CB, and under what conditions — ? 3. What does "reset" mean procedurally — ? 4. Where must extreme caution be exercised — ? AFM 9 (p 2-1-15), Section 3 General (p 3-1-1)	1. Do NOT reset CBs in flight, EXCEPT when required by a procedural step in Section 3, 3A or 4, or if the PIC judges the reset necessary for safe completion of the flight. 2. Only ONE attempt, if the PIC determines the system is needed for safe completion of that flight, after at least ONE MINUTE has elapsed since the trip, and only if there is no remaining smoke or burning smell. 3. Reset = OPEN (pull out) the CB, wait approximately 2 seconds, then CLOSE (push in). If a CB is found already open, reset means simply close it. 4. Fuel pumps and/or fuel quantity indication systems — arcing might ignite fuel or fuel vapors.

Emergency knowledge

QUESTION	ANSWER
AFM-49 DUAL ENGINE FAILURE — GLIDE SPEED Both engines have failed. Per the AFM WARNING, what speed is recommended with flaps up, and what protection is lost? AFM 3-ENG-01 (p 3-13-1)	WARNING: AVOID STALL — SHAKER AND PUSHER ARE UNAVAILABLE. WITH FLAPS UP, 150 KIAS IS RECOMMENDED AS A SAFE SPEED WITH GOOD GLIDE PERFORMANCE. Procedure order (3-ENG-01): 1. Airspeed — As required · 2. Cabin Altitude — Monitor · 3. Emergency Descent 3-NAE-02 — Accomplish · at or below 25,000 ft: 4. Engine Airstart 3-NAE-03 — Accomplish · if no airstart successful: 5. Forced Landing 3-NAE-15 CAUTION: Start with the engine deemed most likely to relight successfully. NOTE: 3-ENG-01 has NO red memory-item box — step 1 is not a boxed memory item.
AFM-50 ENGINE AIRSTART — STARTER ASSIST LIMITS For a starter-assisted airstart: 1. Battery voltage to confirm — ? 2. Airspeed limit if only battery power is available — ? 3. N2 required before selecting RUN — ? 4. When must you not airstart at all — ? AFM 3-NAE-03 (p 3-3-3)	1. BAT 2 — Confirm ON / above 23.0 V. If <23.0 V, consider a windmill airstart. 2. CAUTION: If ONLY battery power is available (after loss of all generated power), only a starter-assisted airstart is permitted, and airspeed must be <=150 KIAS (ensuring N2 <10%). Attempting a starter-assisted airstart above 150 KIAS may cause loss of BOTH FADEC channels. 3. ENG switch — check N2 below 24%, then select RUN. 4. Do NOT airstart if engine integrity is in question or after an engine fire.

QUESTION	ANSWER
AFM-51 ENGINE FIRE — SECOND EXTINGUISHER After firing the first extinguisher bottle and the fire CAS message remains: 1. What confirms the second bottle is available — ? 2. May the engine be restarted after a confirmed fire — ? AFM 3-FIRE-01 (p 3-15-1)	1. CONFIRM green "2" illuminated approximately 30 sec after firing the first extinguisher, then push the associated EXTINGUISHER push button. 2. CAUTION: Do NOT restart the engine after a confirmed engine fire. Also: BLEED Rotary Selector — select operating engine side; FUEL — monitor balance; FUEL X-FEED should be closed prior to landing; Aircraft — land as soon as possible.
AFM-52 LANDING — NOSE LANDING GEAR UNLOCKED Nose L/G indicates unlocked. State the key actions. AFM 3-NAE-18 (p 3-6-x)	1. L/G position — attempt to confirm by exterior means 2. CPCS DUMP pushbutton — when below 10,000 ft AMSL, push to DUMP 3. Landing configuration — FLAPS 33° 4. Aircraft — touch down with normal landing attitude, gently lower the nose to the ground while the elevator remains effective; do NOT use the brakes until the nose gear is on the ground If L/G collapses during touchdown or if considered appropriate — accomplish Emergency Evacuation 3-EVAC-01. NOTE: Consider pressing AURAL DISABLE to suppress "Gear Gear" and TAWS callouts.

General ops

QUESTION	ANSWER
AFM-53 "LAND AS SOON AS POSSIBLE" VS "PRACTICAL" Define the two AFM landing urgency terms. AFM Section 3 General (p 3-1-1)	LAND AS SOON AS POSSIBLE — land without delay at the nearest airport where a safe approach and landing is reasonably assured. LAND AS SOON AS PRACTICAL — the landing airport and duration of flight are at the discretion of the pilot. Extended flight beyond the nearest suitable airport is not recommended.
AFM-54 MEMORY ITEMS — DEFINITION AND MARKING How does the AFM define a memory item, and how are memory items marked? AFM Section 3 General §3, Section 3A General §3	A memory item is a check for an abnormal or emergency situation that requires IMMEDIATE action and is therefore carried out from memory, without prior reference to a checklist. Memory items are executed by heart without referring to a checklist. A specific checklist may be required to complete the action if the memory items cover only the initial actions. Memory items are denoted by a SOLID RED BOX around specific challenge and response items.
AFM-55 MINIMUM FLIGHT CREW AND KINDS OF OPERATION 1. Minimum required flight crew — ? 2. Certification category — ? 3. Approved kinds of operation — ? AFM 12, 17 (p 2-1-17, 2-1-22)	1. One pilot in the LEFT HAND seat. 2. Commuter Category (approved based on CS-23 through Amendment 3). 3. VFR Day · VFR Night · IFR Day including automatic approaches to CAT I minimums, single pilot · IFR Night including automatic approaches to CAT I minimums, single pilot · Flight Into Known Icing (FIKI). All installed equipment shall be operative at dispatch (normally indicated by the absence of CAS messages) unless an approved (M)MEL is used.
AFM-56 REQUIRED DOCUMENTATION ON BOARD Which documents must be aboard in PAPER, and which may be electronic or paper? AFM 2 (p 2-1-1)	PAPER (hard copy), readily accessible to the pilot: • PC-24 Airplane Flight Manual (AFM) Report No. 02371, Cockpit Handbook Volumes 1 and 2 ELECTRONIC (digital) OR PAPER, for reference: • PC-24 Quick Reference Handbook (QRH) Report No. 02382 • Advanced Cockpit Environment (ACE) Avionics System Pilot's Guide applicable to that serial number's software/hardware configuration • PC-24 Flight Crew Operating Manual (Document No. 02383) The AFM must be carried in the airplane at all times.
AFM-57 WET RUNWAY — DEFINITION How does the AFM define a wet runway? AFM 3 (p 2-1-1)	A runway whose surface is covered with a layer of water LESS THAN 1/8 in (3 mm) in depth, or the equivalent amount of a related substance, or has a sufficient level of moisture to give a REFLECTIVE APPEARANCE — but WITHOUT any significant area of standing water. Approved surfaces: dry and wet paved runways. Contaminated paved runways per AFM Supplement 02442. With the gravel kit (stone guard) installed: dry/wet prepared dirt-sand-gravel per Supplement 02444, dry/wet prepared grass per Supplement 02473.

QUESTION	ANSWER
AFM-58 EXTENDED TEMPERATURE ENVELOPE — MINIMUM MACH For the extended operating temperature envelope (OAT < -65 °C), state the minimum operating speed for each temperature. AFM 22.2 (p 2-1-26)	-66 °C — 0.59 M -67 °C — 0.62 M -68 °C — 0.65 M -69 °C — 0.68 M -70 °C — 0.70 M -71 °C — 0.73 M The minimum operating speed provides adequate aerothermodynamic heating to maintain minimum skin and local ambient (zone) temperatures. Sea level limits: min OAT -54 °C (-65 °F) · max OAT +50 °C (122 °F). Max operating altitude 45,000 ft (13,716 m).
AFM-59 RVSM — REQUIRED EQUIPMENT AND CRITICAL AREA 1. What equipment must be operational to enter RVSM airspace — ? 2. What are the dimensions of the RVSM critical area — ? 3. May the ESIS be used for RVSM — ? AFM 25 (p 2-1-37, 2-1-38)	1. Two primary Air Data Systems (ADS) · one flight controller KMC 9200A with altitude pre-selector · one automatic flight control system · one altitude reporting transponder KXP 2290A. 2. An area 24 in (0.6 m) FORWARD, and 12 in (0.3 m) above, below and AFT of the pitot-static tube — must be free of paint ridges/flaking, dents, skin deformation/delamination, loose or missing rivets and fasteners, nicks, scoring/scratches and corrosion. Nose doors closed and latched, panel edges flush. 3. No. The ESIS has not been demonstrated to meet RVSM performance requirements and shall only be used for emergency procedures.
AFM-60 FMS — CONDITIONS FOR INSTRUMENT APPROACHES Use of the FMS to conduct instrument approaches is permitted provided what five conditions are met? AFM 4.3.1, 4.3.2 (p 2-1-4)	• The reference coordinate datum system for the approach is WGS-84, AND • The approach is an approved instrument approach procedure, AND • The approach is retrieved from the FMS database, AND • APP (approach active) mode is annunciated at the Final Approach Fix, AND • The approach is NOT any of these prohibited types: ILS, LOC, LOC-BC, LDA, SDF, MLS FMS use for IFR requires the most current database update cycle. The PIC must verify FMS-managed speeds do not violate published procedural speeds.
AFM-61 VNAV / LNAV APPROACH LIMITATIONS 1. What must remain the primary altitude reference when using VNAV — ? 2. Which minimums are controlling — ? 3. What is the CAUTION about the VNAV deviation indicator — ? AFM 4.3.3 (p 2-1-4)	1. The BAROMETRIC ALTIMETER must be used as the primary altitude reference at all times. 2. The published LNAV/VNAV minimums on the approach chart are controlling; all operations below them must be conducted by outside visual reference. On an LNAV approach using FMS vertical guidance, never descend below the published LNAV MDA unless the required runway visibility exists. 3. CAUTION: Due to the large tolerances of the Vertical Navigation system, the deviation indicator must NOT be relied on when operating below the applicable published minimum.
AFM-62 WATER WASTE SYSTEM LIMITATION When must the water waste system not be operated or serviced? AFM 28 (p 2-1-41)	If installed, do NOT operate the Water Waste System in flight or on the ground if the cabin temperature is ≤2 °C (≤36 °F). Do NOT service the system in freezing conditions — if servicing is required, it must be done in a heated hangar.
AFM-63 PORTABLE ELECTRONIC DEVICES What are the limitations on passenger electronic device and wireless use? AFM 27.1 (p 2-1-40)	Use of Portable Electronic Devices is permitted during ALL phases of flight. There are NO limitations on the use of wireless systems, including Wi-Fi, Bluetooth and cellular systems (GSM, UMTS/3G, LTE/4G).
AFM-64 DYNAMIC SPEED BUG AND STALL WARNING — ADVISORY ONLY What is the status of the Dynamic Speed Bug, low speed awareness and stall warning systems, and what must you rely on instead? AFM 4.1, 4.2.1 (p 2-1-2)	They are for ADVISORY USE ONLY. Maintain a safe stall margin based on the Stall Speed charts in Section 5 (Performance) of the AFM. NOTE: Use of "PITCH ATTITUDE HOLD" mode is recommended during operation in severe turbulence.

QUESTION	ANSWER
AFM-65 TOLD — ADVISORY STATUS AND EXCLUSIONS 1. What is the status of TOLD calculations — ? 2. Name the four Section 5 exceptions TOLD does not cover. 3. Does TOLD consider terrain for climb performance — ? AFM 4.4 (p 2-1-6)	1. ADVISORY PURPOSES ONLY — it must always be cross-referenced against the AFM or other certified performance calculation software. 2. (a) Acceleration and final segment for OEI climb procedure (b) consideration of turns in the OEI climb procedure for obstacle clearance (c) cold temperature compensation below standard temperatures (d) abnormal landing scenarios such as system failures and emergency landings. 3. No — TOLD does not consider terrain data for climb performance. It is the operator's responsibility to determine the most limiting obstacle. TOLD does not support Supplements 02442, 02444, 02451, 02457 or 02473.

ACE avionics

Working knowledge — alerting, mode logic, reversion.

Alerting & aural

QUESTION	ANSWER
ACE-1 CAS MESSAGE LEVELS — COLOUR, LIGHT AND CHIME For each CAS level state the colour, whether a master light illuminates, and the aural: 1. Warning — ? 2. Caution — ? 3. Advisory — ? 4. Status — ? ACE Pilot's Guide 8-6 to 8-8	1. WARNING — RED · master WARNING button lights · TRIPLE CHIME · posted in red reverse video until a master warning button is pushed, then red text on black 2. CAUTION — AMBER · master CAUTION button lights · SINGLE CHIME · amber reverse video until acknowledged, then amber text on black 3. ADVISORY — CYAN · no light, no chime · cyan reverse video for 5 seconds, then automatically to cyan on black 4. STATUS — WHITE · no light, no chime · white reverse video for 5 seconds, then automatically to white on black Pushing a master button silences the chime and drops the reverse video — that is what "acknowledged" means.
ACE-2 CAS WINDOW — SORT ORDER AND CAPACITY 1. In what order are CAS messages sorted — ? 2. How are messages ordered within a colour group — ? 3. How many messages display at once — ? 4. Can the pilot clear a message — ? ACE Pilot's Guide 8-2 to 8-8	1. RED at the top, then AMBER, then CYAN, then WHITE 2. Newest at the top of each colour group, oldest at the bottom 3. As many as 12 at one time. With 12 or fewer, no manipulation is needed 4. NO — the pilot cannot directly remove a message from the CAS stack. A message stays until the condition clears (or the CAS system removes it on a configuration change) When no messages are active, "CAS" is displayed on the top line. The CAS window is a high-priority window and is ALWAYS visible — on a composite PFD it moves to the bottom of the display, below the HSI.
ACE-3 AURAL WARNING PRIORITY — HOW CONFLICTS RESOLVE Two or more aural warnings are active at once. How does the system decide what you hear, and what is always first? ACE Pilot's Guide 8-36, Table 8-5	Aurals are assigned to GROUPS by importance and urgency, and given a PRIORITY NUMBER within the group. ALL warnings from the most important group are presented first, in priority order. Warnings from the next group are not presented until no warnings from the more important group remain active. STALL is group 1, priority 1 — it outranks everything and is NOT mutable. It persists until the stall condition is corrected. Worked example from the guide: with stall, speed and fire warnings all active, you hear STALL first (and only STALL) until the stall is resolved, then the speed warning, then the fire warning.
ACE-4 AURAL PRIORITY ORDER — THE TOP OF THE LIST State the aural message and relative priority for: stall, gear, overspeed, takeoff configuration, speed brake, cabin pressurization, and a red CAS warning. ACE Pilot's Guide Table 8-5 (8-37)	Priority 1 — STALL (continuous, not mutable) Priority 5 — "GEAR" (continuous, not mutable) Priority 6 — "SPEED" — overspeed (continuous, not mutable) Priority 7 — "CAT 2 FAIL" (single) Priority 8 — "NO TAKEOFF" — takeoff configuration (continuous, not mutable) Priority 9 — "SPEED BRAKE" (continuous, not mutable) Priority 10 — "CABIN" — cabin pressurized (continuous, not mutable) Priority 11 — TRIPLE CHIME — CAS warning (continuous, mutable) Note the pattern: the un-mutable, continuous aurals are the ones tied to an immediate flight-path or configuration hazard.

QUESTION	ANSWER
ACE-5 AURAL PRIORITY ORDER — FIRE, CABIN AND TRIM State the aural and priority for: pitch/yaw trim runaway, pitch trim in motion, left and right engine fire, cabin altitude, emergency descent, battery hot. ACE Pilot's Guide Table 8-5 (8-37, 8-38)	Priority 12 — "Trim Runaway" — pitch trim runaway (continuous, mutable) Priority 13 — "Trim Runaway" — yaw trim runaway (continuous, mutable) Priority 14 — CLACKER — pitch trim in motion (continuous, mutable) Priority 15 — "LEFT ENGINE FIRE" (continuous, mutable) Priority 16 — "RIGHT ENGINE FIRE" (continuous, mutable) Priority 17 — "CABIN ALTITUDE" (continuous, mutable) Priority 18 — "EMERGENCY DESCENT" (single, not mutable) Priority 19 — "Battery Hot" (continuous, mutable)
ACE-6 AURAL PRIORITY ORDER — AUTOMATION AND ADVISORY State the aural and priority for: AT engaged, CAS caution, AP uncommanded disconnect, minimums, AP commanded disconnect, altitude alert, autothrottle disconnect, ice mode. ACE Pilot's Guide Table 8-5 (8-38)	Priority 23 — "AT ENGAGED" (single) Priority 24 — SINGLE CHIME — CAS caution (continuous, mutable) Priority 25 — CAVALRY CHARGE — AP UNCOMMANDED disconnect (CONTINUOUS) Priority 26 — "MINIMUMS" (single) Priority 27 — CAVALRY CHARGE — AP COMMANDED disconnect (SINGLE) Priority 28 — C CHORD — altitude alert (single) Priority 29 — C CHORD — vertical track alert (single) Priority 30 — "AUTOTHROTTLE" — AT abnormal disconnect (continuous, mutable) Priority 31 — "AUTOTHROTTLE" — AT normal disconnect (single) Priority 32 — "ICE" — ice mode alert (single) The commanded/uncommanded distinction is the one that matters: same cavalry charge, but CONTINUOUS means the aircraft dropped the autopilot, not you.
ACE-7 SILENCING THE CAVALRY CHARGE An uncommanded AP disconnect is sounding. How is that specific tone muted, and what will NOT mute it? ACE Pilot's Guide Table 8-5 Note 3 (8-39)	The AP uncommanded disconnect tone is mutable ONLY via: • an AP QUICK DISCONNECT button (either yoke), or • the AP button on the Flight Controller Pushing Master Caution or Master Warning will NOT silence it — those mute other aural, not this one. This is why the QRH AFCS Abnormal Disconnect procedure has you grasp the wheel and then press the AP disengage switch: that press is what stops the noise.
ACE-8 AURAL ROUTING — EGPWS AND TCAS VS CAS 1. How do EGPWS and TCAS aural reach you, relative to CAS aural — ? 2. What happens when a TAWS alert and a TCAS alert coincide — ? 3. Which datalink aural are inhibited, and when — ? ACE Pilot's Guide Table 8-5 Notes 1, 2, 4, 5 (8-39)	1. EGPWS and TCAS aural are input DIRECTLY to the Audio Panel and come straight to the headset, so they are heard SIMULTANEOUSLY with CAS-generated aural. 2. All TAWS alerts MUTE the TAS/TCAS alerts — terrain wins over traffic. 3. ATC datalink aural are inhibited during engine start, takeoff and landing; any alert received is annunciated at the completion of that phase. The SATCOM aural is inhibited during engine start, takeoff, approach and landing roll.

AFCS & modes

QUESTION	ANSWER
ACE-9 FLIGHT MODE ANNUNCIATOR — LAYOUT AND COLOUR LOGIC 1. Where are lateral vs vertical FD modes displayed — ? 2. What colour is an ACTIVE mode? An ARMED mode — ? 3. What happens visually when an armed mode captures — ? ACE Pilot's Guide 6-17, 6-18	1. Lateral modes are displayed to the LEFT of the AP engage annunciator; vertical modes to the RIGHT of the YD annunciator. The FMA sits in the upper-middle of the PFD. 2. ACTIVE = GREEN. ARMED = WHITE. 3. On capture the armed mode transitions to active and FLASHES GREEN REVERSE VIDEO for approximately 5 seconds, then remains steady green. FD modes are removed when the AP is not engaged and the FD has been deactivated.
ACE-10 AUTOPILOT STATUS ANNUNCIATOR State the meaning of each AP annunciator: 1. Green AP — ? 2. Amber TCS — ? 3. Red AP flashing 5 seconds — ? 4. Red AP flashing continuously — ? ACE Pilot's Guide Table 4-25 (4-57)	1. Green AP — autopilot engaged 2. Amber TCS — the TCS button is pushed 3. Red AP flashing for 5 seconds, then out — NORMAL disengagement. No pilot acknowledgment required. 4. Red AP flashing CONTINUOUSLY until acknowledged with the Quick Disconnect — a FAILURE occurred while engaged. An "AP X Fail" advisory is displayed while the failure is present. Only one AP annunciator can be displayed at a time. Continuous = the aircraft dropped it; brief = you commanded it.

QUESTION	ANSWER
ACE-11 AUTOPILOT ENGAGEMENT AND DISENGAGEMENT 1. What are the three normal means to manually disengage the AP — ? 2. What does engaging the AP also engage — ? 3. What happens if you engage the AP with the FD off — ? ACE Pilot's Guide 4-56, 4-58	1. • the AP button on the Flight Controller • an AP quick-disconnect button • activation of manual stabilizer trim 2. Selecting AP also activates the PITCH AUTOTRIM function and the YAW DAMPER (when not already active). Manual deselection of the AP does NOT deselect the YD. 3. The FD automatically turns ON, with PITCH HOLD as the active vertical mode and ROLL HOLD as the active lateral mode. The AP can also be disengaged with the TCS or QD buttons. A yaw damper failure due to YD-only faults does NOT disengage the autopilot.
ACE-12 YAW DAMPER — INTERLOCKS YOU MUST KNOW 1. What happens to the AP when you disengage the YD — ? 2. Above 30,000 ft, how must the YD be disengaged — ? 3. When does the YD disengage automatically — ? 4. Green vs amber YD — ? ACE Pilot's Guide 4-63, Tables 4-28, 4-29	1. Disengaging the YD ALSO disconnects the AP. (The reverse is not true: deselecting the AP with the AP button or manual electric trim does not disengage the YD.) 2. The YD is INHIBITED from disengaging above 30,000 ft using the QD button — it must be disengaged with the YD button on the Flight Controller. 3. Automatically when radio altitude is valid and the aircraft transitions below 30 FEET. 4. Green YD = engaged. Amber YD = engaged, then a failure occurred. Pushing either the YD button or the QD button gives a red AP flashing for 2.5 sec plus the AP disconnect tone; no acknowledgment required.
ACE-13 AUTOPILOT CONTROL AND OVERPOWER LIMITS 1. AP roll and pitch control limits — ? 2. Maximum momentary overpower force at the column, pitch and roll — ? 3. Roll and pitch acceleration limits — ? ACE Pilot's Guide 4-55, 4-58	1. Roll ± 35 degrees · Pitch ± 20 degrees 2. Not more than 50 POUNDS in pitch and 30 POUNDS in roll, measured at the control column 3. Roll acceleration limited to $+6 \text{ deg/s}^2$ · pitch acceleration limited to $+3.0 \text{ deg/s}^2$ If the AP is engaged while outside the pitch or roll limits, it follows FD commands to bring the aircraft back within limits.
ACE-14 TOUCH CONTROL STEERING (TCS) What does pressing and releasing TCS do with the autopilot engaged? ACE Pilot's Guide 4-56	PRESSING TCS: the AFCS neutralizes the autopilot by RELEASING the aileron and elevator servo CLUTCHES, letting you hand-fly. An amber TCS is annunciated. RELEASING TCS: the autopilot clutches re-engage. The flight director modes react as if the autopilot had just been engaged — targets are SYNCHRONIZED with the current flight conditions. So TCS is the tool for a quick manual correction without dropping the autopilot — but be aware your targets resync on release.
ACE-15 THRUST RATING SELECTION — AUTO-RATING TYPES In auto-rating mode, what are the rating types, and how many can be active at once? ACE Pilot's Guide 4-81, Table 4-32	Four auto-rating types: TAKEOFF (TO) · MAX CONTINUOUS THRUST (MCT) · CLIMB (CLB) · CRUISE (CRZ). Rating types are MUTUALLY EXCLUSIVE — only ONE engine data rating type is active at a time. (IDL and ATR are also rating types.) The TRS picks the rating from the phase of flight and communicates the active rating to the displays and to the autothrottle for scheduling engine response. Transitions back to TAKEOFF require: FD mode PIT, (V)FLC, VS, (V)ASEL, (V)ALT or VPATH, both engines on, and landing gear NOT extended. Any of flaps in takeoff config, gear extended, or FD mode GA drives it out of CLB/CRZ.

Automation protections

QUESTION	ANSWER
ACE-16 EMERGENCY DESCENT MODE — ACTIVATION What three conditions must all be present for EDM to activate? ACE Pilot's Guide 4-69	ALL of: 1. A cabin DEPRESSURIZATION is detected 2. Aircraft altitude is greater than 30,000 FEET 3. The AUTOPILOT is ON EDM is an OPTION and must be enabled via APM to operate. Both the lateral AND vertical fields of the FMA display an amber EDM to tell you the system is active.

QUESTION	ANSWER
ACE-17 EMERGENCY DESCENT MODE — WHAT THE AIRCRAFT DOES EDM has activated. Describe the full sequence the aircraft flies. ACE Pilot's Guide 4-69	1. Performs a 90° LEFT turn 2. ENGAGES the autothrottle 3. Commands thrust to IDLE 4. Descends at VMO to 15,000 FEET 5. At 15,000 ft: altitude capture, then the AP transitions to HEADING mode and 175-KNOT SPEED HOLD mode Know the left turn and the 15,000 ft / 175 kt end state — that is what you will be asked to predict and to monitor.
ACE-18 EMERGENCY DESCENT MODE — CANCELLING IT 1. What four actions cancel EDM — ? 2. What do the other Flight Controller buttons do — ? 3. Does the AP QD button disconnect the autothrottle — ? ACE Pilot's Guide 4-70	1. • the AP QD button on the control wheels • activation of MANUAL TRIM • the AP button on the Flight Controller • the TCS button 2. ALL other Flight Controller buttons are IGNORED and have no effect while EDM is active. 3. NO. Pushing the AP QD button during EDM does NOT disconnect the AT. The AT must be disconnected with the AT button on the Flight Controller or the AT QD button on the thrust lever. On releasing TCS, the AFCS reverts to basic PIT and ROL modes.
ACE-19 SWPS — WARNING, PUSHER, AND HOW TO STOP IT 1. How does the stall warning present — ? 2. What does the stall warning function do to the autopilot — ? 3. How is an active stick pusher disconnected — ? ACE Pilot's Guide 4-70, 4-71	1. STALL / STALL* annunciation on the PFD, the aural "STALL", and STICK SHAKER activation. 2. The stall warning function DISENGAGES THE AUTOPILOT before reaching the stick pusher condition. 3. By HOLDING the QD button on the pilot's yoke. (The pusher activates when the pilot does not respond to the stall warning — it engages the pitch servo and noses the aircraft down until the stall condition has passed.) The SWPS computes from TWO AOA VANES, one each side, plus FLAPS and ICE MODE inputs. Left shaker/left AOA drives the pilot-side LSA; right drives the copilot-side LSA.
ACE-20 SWPS CAS MESSAGES AND THEIR LEVEL State the alert level for: Pusher Fail, Shaker Fail, and L/R LSA Fail. ACE Pilot's Guide Table 4-31 (4-71)	Pusher Fail — CAUTION. At least one instance of the pusher is not available for engagement. Shaker Fail — CAUTION. At least one instance of the shaker is not available. L LSA Fail / R LSA Fail / L+R LSA Fail — ADVISORY. The low speed awareness "thermometer" on that PFD has failed and will not be displayed. Remember the AFM warning that pairs with Pusher Fail: natural stalls are NOT prevented with the pusher inoperative — stalls must be avoided.
ACE-21 TACTILE FEEDBACK SYSTEM — BANK AND OVERSPEED 1. When is TFS active — ? 2. At what bank does it engage, and until what — ? 3. How do those numbers change in an overspeed condition — ? ACE Pilot's Guide 4-68, 4-69	1. TFS is active when the aircraft is IN THE AIR and the AUTOPILOT IS OFF. (Option.) 2. If the aircraft banks through 51°, the roll servo engages and opposes the pilot. The opposing force continues until bank is reduced to 31°. 3. When TFS is activated for an OVERSPEED condition, bank protection activates at 31° instead of 51°, and opposes until bank is reduced to 11°. BNK and TF are displayed in the FMA; TF flashes reverse video until disengaged. Overspeed protection targets VMO/MMO — NOT placard speeds.
ACE-22 TACTILE FEEDBACK SYSTEM — OVERRIDING IT How do you override or disconnect the TFS, and what is the difference between the two methods? ACE Pilot's Guide 4-68	AP QD button (either yoke): DEACTIVATES the TFS and INHIBITS it from reactivating until conditions are met where TFS would again be active. TCS button: overrides the opposing force TEMPORARILY. When TCS is released, the TFS RE-ENGAGES if the aircraft is still within the conditions where TFS would be active. So QD is the "make it stop" action; TCS is the "hold it off while I maneuver" action.
ACE-23 RUDDER BIAS 1. When is the rudder bias system active — ? 2. What two conditions engage it — ? 3. Which way does it push, and how far — ? 4. What annunciates — ? ACE Pilot's Guide 4-67	1. During the TAKEOFF and GO-AROUND phases, with the YAW DAMPER OFF, up to 2,500 FEET AGL. 2. • the yaw damper is not active, AND • the thrust (N1) difference between the two engines exceeds a defined threshold 3. It commands the rudder TOWARD THE SIDE OF THE HIGHER-THRUST ENGINE at a predefined torque. It is NOT full authority — it supplies only a percentage of the required rudder input through the yaw servo. 4. The RB annunciator REPLACES the yaw damper annunciator in the FMA field. Amber RB means rudder bias is NOT AVAILABLE.

QUESTION	ANSWER
ACE-24 MACH TRIM AND AUTOMATIC PITCH TRIM 1. What does the Mach Trim function do, and when — ? 2. What does automatic pitch trim do, and which motor does it use — ? ACE Pilot's Guide 4-66, 4-67	1. Mach Trim automatically compensates for MACH TUCK by commanding the pitch trim actuator at high Mach numbers WHEN THE AUTOPILOT IS OFF. 2. Automatic pitch trim commands the ALTERNATE STABILIZER TRIM MOTOR to offload steady-state loads on the AP elevator servo — keeping the elevator near neutral, improving cruise performance, and minimizing transients at AP disengagement. Note the split: Mach Trim works with the AP OFF; automatic pitch trim works with the AP ON (it is enabled by AP engagement).
ACE-25 AUTOTHROTTLE — TAKEOFF WINDOW PROTECTION 1. Between what limits does the AT takeoff monitor protect against inadvertent thrust reduction — ? 2. Which lane performs it — ? 3. How does the AT engage into takeoff thrust mode — ? ACE Pilot's Guide 4-73, 4-74	1. From 60 KNOTS to 400 FEET AGL — the "takeoff window". 2. LANE B of the autothrottle performs the takeoff monitor function and controls power to the TQA servos, and can disable AT control of the TQA so the system can reliably be disconnected and inhibited. (The AT function requires two AIOps — lanes A and B. AT is hosted in MAU1 only.) 3. With the AT armed for takeoff, ADVANCING the TLA of BOTH thrust levers while airspeed is LESS THAN 60 KNOTS engages the takeoff thrust control mode.
ACE-26 AUTOTHROTTLE — MODES, PROTECTIONS AND LIMITS 1. What are the primary AT modes — ? 2. Does the AT protect against engine overspeed or over-temperature — ? 3. When does the AT engage automatically — ? ACE Pilot's Guide 4-73, 4-74	1. SPEED, THRUST, LIM and EDM. 2. NO. Engine overspeed and over-temperature protection are FADEC functions, not AT functions. However, the AT ALWAYS HONORS the active engine thrust rating. 3. Automatically when EMERGENCY DESCENT MODE or AUTOTHROTTLE SPEED PROTECTION mode becomes active. The AT mode is synchronous with the active FD pitch mode: thrust mode when speed is being controlled by the FD mode or the FD mode is TO/GA, otherwise speed mode. Disengagement sounds the "AUTOTHROTTLE" aural.

Display failure & reversion

QUESTION	ANSWER
ACE-27 WHICH AGM DRIVES WHICH DISPLAY State the display architecture: which AGM drives which DUs, and how many unique images can one AGM produce? ACE Pilot's Guide 5-48, 5-49	AGM 1 drives DU 1 and DU 2. AGM 2 drives DU 3 and DU 4. DU 1 = pilot PFD · DU 2 and DU 3 = the two middle MFDs · DU 4 = copilot PFD. An individual AGM can generate only TWO different screen images at any one time. That is why, after an AGM failure, the copilot PFD shows an EXACT DUPLICATE of the pilot PFD rather than its own independent image — the images are identical, not symmetrical.
ACE-28 DU FAILURE VS AGM FAILURE — THE CRITICAL DISTINCTION The guide says distinguishing a DU failure from an AGM failure is essential. How do you tell them apart? ACE Pilot's Guide 5-47, 5-49, 5-50	DU FAILURE — that DU goes BLANK (dark, no image), shows a GROSSLY DISTORTED image, or is physically damaged (cracked/shattered). The other three DUs keep working normally; both AGMs are healthy. AGM FAILURE — the two DUs driven by that AGM each display a LARGE "X". A large X means the DU is OPERATING but is NOT RECEIVING IMAGE DATA from the selected AGM. So: X on both DU 1 and DU 2 (or both DU 3 and DU 4) is almost certainly ONE AGM failure — NOT two failed displays. Concluding "two DUs failed" would be wrong, precisely because they are displaying the X.
ACE-29 SINGLE DU FAILURE — ACTION A single DU has failed. What is the corrective action? ACE Pilot's Guide 5-47, 5-49	Turn that DU's DIMMER CONTROL past the detent to the OFF / REV position. This tells the ACE system the display is no longer of any use so it can redistribute the images. The remaining three DUs continue to present all information. Nothing is wrong with either AGM or any other avionics — only that one KDU-1200 display has failed.

QUESTION	ANSWER
ACE-30 AGM FAILURE — ACTION DUs 1 and 2 are both displaying a large X. State the corrective actions in order. ACE Pilot's Guide 5-49	1. Turn the CIRCULAR COLLAR surrounding the PFD DIMMER on the AFFECTED SIDE to select the ALTERNATE (non-normal) AGM to drive the displays. For DUs 1 and 2 showing X, move the collar around the AGM2 dimmer to PILOTS PFD. 2. Then indicate that DU 2 is still showing an X and is of no use: turn the DU 2 dimmer PAST THE DETENT to OFF / REV. 3. Use the TSC MFD SWAP button to alternate between the Situational Awareness and Systems displays on DU 3. What you lose: the copilot no longer has an independently configurable PFD (it is a slave image of the pilot's), and you can no longer show Situational and Systems at the same time.
ACE-31 TWO DUS FAILED — REVERSION LOGIC Two DUs have failed and been turned off. What appears where, for each pair? ACE Pilot's Guide 5-50, 5-51, 5-52	DU 2 + DU 3 (both middle) — COMPOSITE display on DU 1; normal PFD continues on DU 4 DU 1 + DU 2 — left PFD on DU 3; single MFD on DU 4 DU 3 + DU 4 — normal left PFD continues on DU 1; single MFD on DU 2 DU 1 + DU 4 (both outboard) — left PFD on DU 2; single MFD on DU 3 In every case the single MFD toggles between Situational Awareness and Systems with the TSC MFD Swap button. The logic is optimized for SINGLE-PILOT operations: the composite display — the hardest to work with — only appears if BOTH MIDDLE displays are gone. In two-crew ops the copilot (full PFD on DU 4) should take pilot-flying duties.
ACE-32 DISPLAY INTEGRITY MONITORS Name the two display monitors and what each one checks. ACE Pilot's Guide 5-45, 5-46, Table 5-1	DU CRC MONITOR — constantly evaluates the display integrity of each DU individually, verifying that image data generated by the AGM is properly rendered on that DU. If tripped, a "Check DU X" CAUTION is posted; that DU is considered UNRELIABLE and should be reverted off (dimming knob to OFF / REV). Inhibited for a DU that is not powered. CRITICAL SYMBOL MONITOR — compares AGM-generated image data against the raw data for critical items, confirming the graphics processor is not outputting corrupt image data. Critical items: airspeed, baro altitude, baro correction, pitch, roll, zero-pitch reference line and flight path symbol on each PFD; N1 fan speed (and status) and ITT (and status) in the engine windows.
ACE-33 CRITICAL SYMBOL MONITOR CAS MESSAGES — ACTIONS State the required action for: "Check Pilot PFD", "Check Copilot PFD", and "Check Engine Display". ACE Pilot's Guide Table 5-2 (5-46)	Check Pilot PFD — cross-check critical data on the pilot PFD against the STANDBY attitude indicator or the copilot PFD. If the data does not reasonably agree, REVERT THE PILOT PFD TO AGM2. Check Copilot PFD — cross-check against the standby AI or the pilot PFD. If it does not reasonably agree, REVERT THE COPILOT PFD TO AGM1. Check Engine Display — cross-check ITT and N2 fan speed against another display. If they do not reasonably agree, revert the affected PFD to an alternate AGM. In every case the drill is the same: CROSS-CHECK FIRST, revert only if the data disagrees.

Data integrity

QUESTION	ANSWER
ACE-34 PFD MISCOMPARE THRESHOLDS State the difference between left and right PFD that triggers each miscompare: 1. IAS? — ? 2. ROLL? — ? 3. PITCH? — ? 4. ALT? — ? 5. BARO? — ? ACE Pilot's Guide 6-10, 6-26, 6-71	1. IAS? — 10 KNOTS or more (upper-left of both airspeed indicators) 2. ROLL? — 6 DEGREES or more (upper-left of both PFDs) 3. PITCH? — 5 DEGREES or more (upper-right of both PFDs) 4. ALT? — 200 FEET or more (upper-right of both altitude displays) 5. BARO? — 0.02 inHg or more (or 1 hPa), when the altimeters are set up for synchronization (lower-right of both altitude displays) All are displayed in AMBER REVERSE VIDEO. Mnemonic for the attitude pair: roll tolerance is the larger number (6), pitch the smaller (5).

QUESTION	ANSWER
ACE-35 RADIO ALTITUDE AND VERTICAL DEVIATION MISCOMPARE 1. How is the RAD? miscompare threshold computed, and how long must it persist — ? 2. What triggers a VDEV? miscompare — ? ACE Pilot's Guide 6-46, 6-56	1. RAD? appears when the difference between radio altitude indications exceeds a predefined threshold for MORE THAN 1 SECOND. The threshold is computed as: $10 + 0.0625 \times [\text{ABS}(\text{Pilot RadAlt}) + \text{ABS}(\text{Copilot RadAlt})]$ — i.e. the allowable difference GROWS with height, so it is tightest near the ground where it matters. 2. VDEV? appears below the vertical deviation scale when the pilot side and copilot side have selected TWO DIFFERENT FMSs as navigation source, the FMS vertical deviation type is VPATH, and the difference between left and right vertical deviation exceeds 50% OF FULL SCALE.
ACE-36 INVALID DATA PRESENTATION How does the PFD present invalid airspeed, altitude, or attitude data — and how does that differ from a miscompare? ACE Pilot's Guide 6-10, 6-71	INVALID data is REMOVED and replaced with a large X over the affected area: - Invalid airspeed — the airspeed tape is removed and replaced with an X; the rolling digits are removed - Invalid altitude — all altitude data removed, X over the entire altitude tape A MISCOMPARE (amber "?" annunciator) means both sides are producing data but they DISAGREE — you must decide which to trust, typically by cross-checking the standby instrument. An X means that data is GONE. Different problem, different action: miscompare -> cross-check and choose; X -> use the other source.

TCAS

QUESTION	ANSWER
ACE-37 TCAS AURAL NOTIFICATIONS State the aural for: a new traffic advisory, a further TA while one is active, an RA that has cleared, and the self-test result. ACE Pilot's Guide Table 15-2 (15-51)	New traffic advisory — "TRAFFIC, TRAFFIC" When a previous TA is already active — "TRAFFIC" RA cleared (TCAS II only) — "CLEAR OF CONFLICT" Self-test passed — "TCAS SYSTEM TEST OK" Self-test failed — "TCAS SYSTEM TEST FAIL"
ACE-38 RESOLUTION ADVISORIES — CORRECTIVE VS PREVENTIVE What is the aural difference between a CORRECTIVE and a PREVENTIVE RA? ACE Pilot's Guide Table 15-3 (15-52)	CORRECTIVE RA — tells you to CHANGE the flight path: Climb — "CLIMB, CLIMB, CLIMB" Descent — "DESCEND, DESCEND, DESCEND" Crossover climb — "CLIMB, CROSSING CLIMB, CLIMB" PREVENTIVE RA — tells you NOT to change what you are doing: "MONITOR VERTICAL SPEED, MONITOR VERTICAL SPEED" A preventive RA requires you to keep out of a vertical band — it does not require a maneuver. RAs are TCAS II installations only.
ACE-39 TRANSPONDER — NEVER SELECT STBY ON THE GROUND Why must the transponder not be set to STBY on the ground, and what should be selected instead? ACE Pilot's Guide 15-57, 15-58	The ACE transponder supplies AUTOMATIC air-to-ground (GND) switching, so the old practice of selecting STBY on the ground does NOT apply. The pilot must NEVER set the active mode to STBY while on the ground. Instead select the mode you want IN FLIGHT — normally TA/RA for TCAS II (or TA for TCAS I). Transmissions begin automatically at takeoff and stop automatically on landing (except replies to specifically addressed interrogations). On the ground with GND displayed and not in STBY, the transponder still replies to interrogations addressed to the aircraft by name, and other traffic can be viewed with TA or TA/RA selected.
ACE-40 TCAS BEFORE TAKEOFF How is TCAS tested and armed before flight, and what does it do while still on the ground? ACE Pilot's Guide 15-58	Test with the TCAS TEST softkey during cockpit preparation. After passing self-test, place TCAS in TA/RA (TCAS II) or TA (TCAS I) by selecting the transponder to TA/RA or TA. While the aircraft is on the ground (per air-ground logic) TCAS does NOT issue advisories — although threat traffic IS displayed in YELLOW. On transition to air mode the GND annunciator is removed, the transponder begins broadcast operation, and TCAS becomes active.

QUESTION	ANSWER
ACE-41 TCAS LIMITATIONS — WHY TRAFFIC MAY NOT APPEAR Name the situations in which an aircraft you can see is not shown by TCAS. ACE Pilot's Guide 15-59, 15-60	• NO OPERATING TRANSPONDER — TCAS cannot detect it at all • NON-ALTITUDE REPORTING (NAR) intruder — range and bearing only; can produce a TA but never an RA; TCAS ASSUMES NAR traffic is CO-ALTITUDE with you • ANTENNA SHADING — most small aircraft have one transponder antenna on the belly; when you are above them the antenna can be shaded so interrogations or replies do not get through. Also happens when the other aircraft maneuvers and blocks line of sight • CONE OF CONFUSION — the upper directional antenna determines bearing only for intruders between -10° and +70° elevation; outside that they are tracked with NO bearing • LOWER MONOPOLE ANTENNA — no bearing available; the intruder is tracked but NOT DISPLAYED unless a TA is issued • A poor transponder on the other aircraft — ground stations have more gain, so ATC may call traffic your TCAS never sees • NO CLOSURE RATE — TCAS derives closest point of approach; co-speed same-direction traffic may never be a threat

EGPWS / TAWS

QUESTION	ANSWER
ACE-42 EGPWS MODE 1 — EXCESSIVE DESCENT RATE 1. When is Mode 1 active — ? 2. Outer boundary alert — ? 3. Inner boundary alert — ? ACE Pilot's Guide 16-31, 16-32	1. Active regardless of flight phase whenever radio altitude is VALID and LESS THAN 2,450 FEET AGL. Lower cutoff at 10 ft RA during landing; re-enabled at 30 ft RA during takeoff. When barometric input is used, cut off at 30 ft and re-enabled at 65 ft. 2. OUTER — GND PROX annunciator and "SINKRATE, SINKRATE". Repeated twice, then silent unless the condition degrades by about 20% (by computed time to impact), which gives two more and repeats the cycle. 3. INNER — PULL UP annunciator and "PULL UP, PULL UP", or optionally a continuous "WHOO-WHOO PULL UP", repeating continuously until the inner boundary is exited. Steep approach selected desensitizes the outer curve by 500 FPM and the inner by 200 FPM.
ACE-43 EGPWS MODE 3 — ALTITUDE LOSS AFTER TAKEOFF 1. When is Mode 3 enabled — ? 2. What is the alert — ? 3. How does the aural repeat — ? ACE Pilot's Guide 16-40	1. Enabled after TAKEOFF or GO-AROUND when the landing gear or flaps are NOT in the landing configuration, and stays enabled until the EGPWS detects the aircraft has gained enough altitude to no longer be in the takeoff phase. 2. A GND PROX caution on the right side of the attitude indicator and the aural "DON'T SINK, DON'T SINK". 3. The aural is annunciated again only when a further 20% OF RADIO ALTITUDE is lost. If that additional altitude is lost, two more aurals are given and another 20% is added into the calculation — ratcheting until the original altitude is recovered. The visual stays on the whole time the envelope is violated.
ACE-44 EGPWS MODE 6 — ADVISORY CALLOUTS List the Mode 6 callouts and the altitude each occurs at. ACE Pilot's Guide Table 16-3 (16-47)	"MINIMUMS" — at minimums "APPROACHING MINIMUMS" — DH/MDA bug setting PLUS 80 FEET "FIVE HUNDRED" — 500 ft AGL "ONE HUNDRED" — 100 ft AGL "FIFTY" — 50 · "FORTY" — 40 · "THIRTY" — 30 · "TWENTY" — 20 · "TEN" — 10 ft AGL Each selected callout is annunciated ONCE PER APPROACH. The system must transition to approach mode to RE-ARM the callouts for the next approach. Valid radio altitude above the highest selected callout must be present; minimums callouts additionally need the landing gear down (or alternate mode 4B select). No visual alerting accompanies Mode 6 callouts.

QUESTION	ANSWER
ACE-45 TERRAIN AND OBSTACLE ALERTS — CAUTION VS WARNING State the aural, colour and repeat behaviour for terrain/obstacle CAUTION and WARNING alerts. ACE Pilot's Guide 16-53, 16-54	CAUTION — "CAUTION TERRAIN, CAUTION TERRAIN" (optionally "TERRAIN AHEAD, TERRAIN AHEAD"); for obstacles "CAUTION OBSTACLE, CAUTION OBSTACLE" (optionally "OBSTACLE AHEAD"). Repeated EVERY 10 SECONDS while inside the caution envelope. Threat terrain is coded AMBER; GND PROX caution on the attitude indicator. WARNING — "TERRAIN, TERRAIN, PULL UP" (optionally "TERRAIN AHEAD, PULL UP"); for obstacles "OBSTACLE, OBSTACLE, PULL UP". "PULL UP" repeats CONTINUOUSLY while inside the warning envelope. Threat terrain is coded RED; PULL UP warning on the attitude indicator. In both cases only terrain along the aircraft track and within $\pm 90^\circ$ OF TRACK is coded, and the threat image appears on the HSI.
ACE-46 EGPWS CLASS A-ONLY MODES Which EGPWS modes are unavailable when Class B TAWS is selected, and what does each do? ACE Pilot's Guide 16-33, 16-34, 16-42, 16-49	MODE 2 — Excessive Closure to Terrain. Alerts on excessive terrain closure rate; the aircraft need NOT be descending — level flight or a climb toward rising terrain can trigger it. Sub-modes 2A and 2B by configuration. In a high-integrity terrain awareness state the maximum altitude is reduced from 2,450 ft to 950 ft to cut nuisance alerts. MODE 4 — Unsafe Terrain Clearance. 4A: cruise/approach, gear not in landing config. 4B: gear in landing config, flaps not. 4C: takeoff phase with either gear or flaps not in landing config. MODE 5 — Excessive Deviation Below Glideslope. Front-course ILS, or below the selected GPS vertical path on approaches to LPV minimums. Needs localizer within ± 2 dots, radio altitude valid and below 1,000 ft, and gear down; below 500 ft it is enabled without a valid localizer. Subsequent Mode 4 aural occur only when envelope penetration increases by 20%.

Verified gouge values

117 lines from a classmate's gouge that were confirmed against the AFM or the FSI Pilot Training Manual. The gouge's wrong, conflicting and unconfirmed lines are **not** reproduced here — see the Gouge tab in the app for those.

Weights — read this first

QUESTION	ANSWER
MAXIMUM BAGGAGE WEIGHT, LARGE RESTRAINT NET Verified: Serial-independent. Small net is 400 lb — the gouge omits it. Card L-4	530 lb
BAGGAGE VS CARGO DEFINITION Verified: PTM App B p771 — baggage is an item with mass > 66 lb (30 kg) classed as cargo. Must be in a restraint system.	Baggage $\leq$ 66 lb, cargo > 66 lb
FLOOR LOADING Verified: Card L-5	100 lb/ft ² over floor panels, 670 lb/ft on seat rails
TAIL STAND Verified: PTM App B p771	Install tail stand before loading cargo

General limitations

QUESTION	ANSWER
MAXIMUM OPERATING ALTITUDE Verified: Card L-7; PTM App B p771	45,000 ft (implied by 'Vo sea level to FL450')
MINIMUM PRESSURE ALTITUDE Verified: Card L-11	-1,000 ft MSL
MAXIMUM PRESSURE ALTITUDE, LANDING FIELD ELEVATION Verified: Card L-11	10,000 ft MSL
MAXIMUM RUNWAY SLOPE Verified: Card L-10; PTM App B p771	$\pm 2\%$

QUESTION	ANSWER
MAXIMUM TAILWIND, TAKEOFF AND LANDING Verified: Card L-11; PTM App B p771	10 kt
MINIMUM / MAXIMUM OAT AT SEA LEVEL Verified: Card L-8	-54 °C / +50 °C (labelled 'Max -Temp / Max +Temp') Values right, labels wrong. These are MINIMUM OAT (-54 °C) and MAXIMUM OAT (+50 °C) at sea level.
LOAD FACTOR, FLAPS UP Verified: Card L-6	+3.0 / -1.2
LOAD FACTOR, FLAPS DOWN Verified: Card L-6	+2.0 / -0.0
TOWBARLESS TUG WEIGHT LIMIT Verified: PTM App B p771 — 5,379 lb (2,440 kg)	5,379 lb

Doors and exits

QUESTION	ANSWER
CABIN DOOR DIMENSIONS Verified: PTM App B p771	2 ft 0.4 in W × 4 ft 5.2 in H
CARGO DOOR DIMENSIONS Verified: PTM App B p771	4 ft 1 in W × 4 ft 3 in H
CARGO DOOR CROSSWIND LIMIT Verified: PTM App B p771	60 kt
AIRSTAIR CAPACITY Verified: PTM App B p771; card L-32	1 person at a time
OVERWING EXIT SECURITY PINS Verified: Card L-33; PTM App B p771	Remove Remove and stow prior to flight.
NOSE GEAR CASTORING / TOWING ANGLE Verified: PTM App B p771	±60° 60°, reached when the red markings on either side of the NLG align.

Speeds

QUESTION	ANSWER
VMO / MMO Verified: Card L-1; PTM App B p774	290 kt / 0.74 M VMO 290 KIAS at or below 28,230 ft; MMO 0.74 M at or above 28,230 ft.
VLO EXTEND Verified: Card L-28; PTM App B p775	Lower of 250 KIAS / 0.74 M
VLO RETRACT Verified: Card L-28; PTM App B p775	200 kt
VLE Verified: Card L-28; PTM App B p775	Lower of 250 KIAS / 0.74 M
VFE FLAPS 8° / 15° Verified: PTM p144 and App B p773	200 kt Confirmed. Note this is NOT on any of your 198 cards — the deck has no VFE card. Worth adding.
VFE FLAPS 33° Verified: PTM p144 and App B p773	175 kt Confirmed. Also missing from your deck.

QUESTION	ANSWER
VO (MANEUVERING), RANGE Verified: AFM p83; PTM App B p774	120–170 kt Correct as a range. It is weight-scheduled: 18,740 lb -> 170 KIAS, 18,300 -> 170, 17,650 -> 165, 17,000 -> 160, 16,000 -> 155, 15,000 -> 150, 14,000 -> 145, 13,000 -> 140, 12,000 -> 135, 11,440 -> 120.
RUDDER — UP TO FL200 Verified: PTM App B p773	130 kt Do not make full or abrupt rudder movements above 130 KIAS up to 20,000 ft.
RUDDER — ABOVE FL200 Verified: PTM App B p773	125 kt Do not make full or abrupt rudder movements above 125 KIAS above 20,000 ft.
RUDDER PEDAL TRAVEL, 1/3 LIMIT Verified: PTM App B p773	185 kt Sea level to 45,000 ft: do not exceed 1/3 pedal travel above 185 KIAS.
RUDDER PEDAL TRAVEL, 1/10 LIMIT Verified: PTM App B p773	290 kt Sea level to 45,000 ft: do not exceed 1/10 pedal travel above 290 KIAS.
VMCG, FLAPS 8°/15° Verified: PTM App B p773	87 kt
VMC TAKEOFF, FLAPS 8° / 15° Verified: PTM App B p773	82 kt / 77 kt Correct at heavy weight only. Flaps 8°: >=13,669 lb -> 82, >12,125 -> 87, >11,023 -> 90. Flaps 15°: >=13,228 lb -> 77, >12,125 -> 81, >11,023 -> 84. Lighter aircraft has a HIGHER VMC.
VMCL LANDING, FLAPS 15° / 33° Verified: AFM p84	78 kt / 73 kt Correct at heavy weight. Flaps 15°: >=13,007 lb -> 78, >12,125 -> 81, >11,023 -> 84. Flaps 33°: >=13,228 lb -> 73, >12,125 -> 77.

Powerplant

QUESTION	ANSWER
ENGINE Verified: Card L-12	Williams International FJ44-4A-QPM, medium bypass ratio
NORMAL TAKEOFF THRUST Verified: Card L-14; PTM App B p776	3,420 lbF
AUTOMATIC THRUST RESERVE Verified: Card L-14; PTM App B p776	3,600 lbF 3,600 lb. The gouge's '10 min OEI, 5 min AEO' timing was not confirmed in the sources checked.
ITT — TAKEOFF / ATR / TRANSIENT HIGH Verified: PTM App B p776	855 °C
ITT — MAX CONTINUOUS / CLIMB Verified: PTM App B p776	835 °C
OIL QUANTITIES Verified: AFM p100	Total 5.85 qt, fill 5.65 qt, usable 4.32 qt Confirmed exactly.
OIL LEVEL CHECK Verified: Card L-13; AFM p100	Keep above ADD mark
ZERO / NEGATIVE G LIMIT Verified: AFM p102	10 sec (oil pressure may indicate 0) Confirmed. The engine is limited to 10 seconds of continuous flight at zero and/or negative G; oil pressure indicating zero is acceptable during that time.

QUESTION	ANSWER
TAILWIND ABOVE 10 KT Verified: PTM App B p776	Do not exceed 60% N1 until groundspeed exceeds the tailwind
ENGINE SHUTDOWN COOLDOWN Verified: Card L-22; PTM App B p776	2 min idle (recommends bleed valves off) Idle >= 2 min with bleed OFF. The AFM states bleed off as part of the limitation, not a recommendation.
GROUND IDLE VALUES Verified: Card E-19	N1 25%, ITT 400 °C, N2 min 53.4%, FF 160–180 lb/hr
ENGINE FIRE PROTECTION Verified: PTM App B p773	2 × 1.5 lb Halon bottles, either can serve either engine Both left and right fire protection systems must be operative for dispatch.

Engine starting

QUESTION	ANSWER
MAXIMUM TAILWIND FOR START Verified: Card L-16; PTM App B p776	20 kt 20 kt estimated steady wind.
MAXIMUM CROSSWIND FOR START Verified: Card L-16; PTM App B p776	25 kt tower wind
MAXIMUM TIME TO LIGHT-OFF Verified: Card L-17; PTM App B p776	12 sec (IGN to ITT rise)
MINIMUM OIL TEMP FOR BATTERY START Verified: Card L-18; PTM App B p776	-40 °C
MINIMUM OIL TEMP, N2 > 80% OR QPM Verified: Card L-19; PTM App B p776	+10 °C
STARTER DUTY CYCLE Verified: Card L-15; PTM App B p776	50 s on / 60 s off ×2, then 50 s on / 30 min off Attempt 1: 50 s on, 1 min off. Dry motor run: 50 s on, 1 min off. Attempt 2: 50 s on, 30 min off.

QPM

QUESTION	ANSWER
GENERATOR LOAD LIMIT IN QPM Verified: Card L-37	250 A
THE 2-MINUTE ANCHOR Verified: Cards L-20, L-21, L-22	Idle 2 min before and after QPM Same 2 minutes governs QPM entry, QPM exit and shutdown cooldown.

Fuel

QUESTION	ANSWER
USABLE CAPACITY Verified: Card L-23	5,964 lb / 890 gal
MAXIMUM PERMISSIBLE IMBALANCE Verified: Card L-23; PTM p370	330 lb / 49 gal
MAXIMUM / MINIMUM FUEL TEMPERATURE Verified: Card L-24	+80 °C / -40 °C
APPROVED FUEL GRADES Verified: Card L-27	Jet A, Jet A-1, JP-8, TS-I TS-1 with a numeral one, not the letter I.

QUESTION	ANSWER
MAXIMUM REFUELING PRESSURE Verified: Card L-25	60 PSI
BOOSTER PUMPS AND CROSSFEED VALVE Verified: Card L-26; PTM App B p773	Must be operative for flight

Electrical

QUESTION	ANSWER
BATTERY CHARGING VOLTAGE CEILING Verified: AFM p90 §11.2 (2-1-16); card AFM-47. Cross-check card L-36 for the 29.5 V GPU figure.	'Charging not to exceed: 32 V' Correct, and worth understanding because it looks like a contradiction and is not. AFM 11.2 Battery Limitations: "The voltage for charging the batteries must not exceed 32 V." That is the battery-side ceiling. The 29.5 VDC figure elsewhere is a different limit — maximum GPU and generator OUTPUT voltage, and max volts to charge from the GPU. The gouge lists both correctly. Card AFM-47 covers this exact trap.
SYSTEM VOLTAGE Verified: consistent with cards L-36/L-37	28 V, left/right split
MAXIMUM VOLTAGE Verified: Card L-37	29.5 VDC
GENERATOR LOAD LIMITS Verified: Card L-37	400 A (250 A in QPM) QPM 250 A, ground idle and above 400 A, flight 400 A, maximum voltage 29.5 VDC.
MINIMUM NI-CAD VOLTS FOR START Verified: Cards L-18, L-38	BAT 1 22.0 V, BAT 2 23.5 V
MINIMUM LI-ION FOR START Verified: Cards L-18, L-38	Green SOC indicators State of charge indicators on BOTH batteries green.
MINIMUM BATTERY TEMP FOR START Verified: Card L-38	Ni-Cad -20 °C, Li-Ion -5 °C
MINIMUM BATTERY TEMP FOR FLIGHT Verified: Card L-38	0 °C
GPU VOLTAGE Verified: Cards L-34, L-35; PTM App B p773	25.0–29.5 VDC, must exceed battery voltage
GPU LOAD CAPACITY Verified: Card L-35; PTM App B p773	1,200 A start / 450 A continuous
GPU MINIMUM VOLTS FOR START Verified: Card L-36; PTM App B p773	24.0 V
GPU VOLTS TO CHARGE BATTERIES Verified: Card L-36; PTM App B p773	28.0 / 29.5 V
MINIMUM OAT FOR GPU START Verified: Card L-36; PTM App B p773	-54 °C

Flight controls

QUESTION	ANSWER
FLAP ALTITUDE LIMIT Verified: PTM App B p773	20,000 ft
LANDING FLAP SETTINGS Verified: PTM App B p773	15° or 33°; no landing at 0° or 8° unless failed

QUESTION	ANSWER
MAXIMUM FLAP IN ICING Verified: PTM App B p773; card L-43 family	15° Do not extend beyond 15° in icing; do not retract below 8° if ice is present on the surface.
SECONDARY STAB TRIM Verified: PTM App B p773	Do not use unless primary has failed

Landing gear, tires, brakes

QUESTION	ANSWER
GEAR OPERATIONAL TEMPERATURE LIMIT Verified: PTM App B p775	-65 °C -65 °C (-85 °F).
EMERGENCY GEAR EXTENSION Verified: PTM App B p775	Handle pulled below 160 KIAS See the VLO EGES conflict under Speeds — 160 KIAS per PTM Appendix B, 180 KIAS per the FSI flash card. Fly 160.
MAXIMUM TIRE SPEED Verified: Card L-29; PTM App B p775	165 kt groundspeed
NOSE GEAR TIRE TYPE Verified: PTM App B p775	Dual chine single tire NLG must be dual chine.
MINIMUM BRAKE PRESSURE FOR TAKEOFF Verified: AFM p503, p511	2,850 psi Confirmed — the Park/Emergency accumulator must be recharged to above 2,850 psi before commencing takeoff.
NO TAKEOFF — ACCUMULATOR THRESHOLD Verified: AFM p513, p514	Park/Emer accumulator less than 2,750 psi Confirmed, and it is a DIFFERENT threshold from the 2,850 psi recharge target. Both numbers are right — 2,850 is what you recharge to, 2,750 is what trips the warning.
BRAKE COOLING Verified: PTM App B p775	120 min after RTO within 20 kt of Vr, or flaps 0° landing, with heavy braking RTO greater than Vr - 20 kt, on ground 120 minutes; also 0° flaps with heavy braking.

Pressurization and environmental

QUESTION	ANSWER
MAXIMUM POSITIVE CABIN DIFFERENTIAL Verified: Card L-31	9.27 psi
MAXIMUM NEGATIVE CABIN DIFFERENTIAL Verified: Card L-31	-0.3 psi
MAXIMUM DIFFERENTIAL FOR TAKEOFF AND LANDING Verified: Card L-31; AFM p136	0.7 psi Confirmed — the cockpit placard reads 'ENSURE CABIN P MAX. = 0.7 PSID BEFORE LANDING'.
NORMAL / PLACARD MAXIMUM CABIN DIFFERENTIAL Verified: AFM p136–138	8.78 psi Confirmed. The cockpit placard reads MAXIMUM CABIN DIFF. PRESS. = 8.78 PSID. Do not confuse it with the 9.27 psid structural limit — the gauge correctly lists both.

QUESTION	ANSWER
CAB LOW DIFFERENTIAL Verified: PTM p543, p548	8.83 psi Confirmed. CAB LOW is a semiautomatic mode that limits maximum differential to 8.83 psi, using destination field elevation as the base reference. It is a selectable mode, not a caution.
CABIN ALTITUDE LIMITER Verified: PTM p550	14,500 ft, even if dumped Confirmed — altitude limiting holds until cabin altitude decreases below 14,500 ft, and DUMP targets 14,500 ft at a rate limited to 35,000 fpm.

Oxygen

QUESTION	ANSWER
CYLINDER QUANTITY Verified: PTM App B p775	77 ft ³ standard, 115 ft ³ optional
MINIMUM QUANTITY FOR DISPATCH Verified: Card L-49; PTM App B p775	680 litres, aviator's oxygen
CREW MASKS Verified: consistent with PTM	2, stored in 100% mode
FIELD HI MODE Verified: PTM App B p775	One pilot must use oxygen Multicrew: one pilot continuous use. Single pilot: continuous use.
PASSENGER MASK ALTITUDE LIMIT Verified: PTM App B p775	40,000 ft
PASSENGER MASK AUTO DEPLOYMENT Verified: PTM p549; App B p775	Normal 12,850 ft, backup 14,800 ft cabin altitude Confirmed. 12,850 ±150 ft via CPCS in all modes; 14,800 ±500 ft triggered by avionics (ACE) as backup.
CREW MASK REMOVAL TEMPERATURE Verified: PTM App B p775	Below -15 °C if overnight Remove temperature 5 °F (-15 °C).

Ice protection

QUESTION	ANSWER
ICE DETECTORS Verified: PTM App B p774	2, only work in the air Confirmed — ice detectors only operate in flight.
NAI ON THE GROUND Verified: Card L-45; PTM App B p776	OAT below 10 °C plus visible moisture, or ice on the nacelle; not available in QPM
NAI IN THE AIR Verified: Card L-46	TAT below 10 °C plus visible moisture, ice crystals, or ice on airframe
DO NOT OPERATE NAI Verified: Card L-47; PTM App B p776	TAT exceeds 10 °C, or in QPM The 10 °C ice line: NAI required below it with moisture, prohibited above it.
WING ANTI-ICE (WAI) INHIBIT Verified: AFM p94 §15.2; PTM App B p773 and p637	System inhibited: on ground (WOW), TAT 15 °C Correct. AFM 15.2: the WAI is inhibited when TAT is above 15 °C or the aircraft is on the ground. Keep it separate from two neighbouring rules: do not OPERATE the WAI above TAT 10 °C unless icing requires it, and windshield heat is inhibited above TAT 10 °C. Three temperatures, three different systems.

QUESTION	ANSWER
FREEZING RAIN / FREEZING DRIZZLE Verified: Card L-43; PTM App B p774	Prohibited Must exit immediately. Operation is not allowed at airports reporting freezing rain or drizzle.
AUTOPILOT IN ICING Verified: Card L-44; PTM App B p774	Periodically disengage; AP off in severe icing Periodically disengage to check for abnormal forces. Do not engage in severe icing.
HOLDING IN ICING ON A SINGLE BLEED SOURCE Verified: consistent with card L-46 family	Prohibited
WING INSPECTION LIGHT Verified: PTM App B p774, p775	Must be operating for known/forecast icing at night
GO-AROUND FLAP RETRACTION IN ICING Verified: PTM App B p773	No more than 8° Do not retract below 8° if ice is present on the surface.

Autopilot and automation

QUESTION	ANSWER
AP DISENGAGE — ALL OPERATIONS EXCEPT APPROACH Verified: Card L-41	Below 1,000 ft AGL
AP DISENGAGE — NON-PRECISION, CIRCLING, VISUAL Verified: Card L-41	Below 400 ft AGL 400 ft AGL, with airspeed below 150 KIAS and vertical speed below 1,500 ft/min.
AP DISENGAGE — VGP MODE Verified: Card L-41	Below 200 ft AGL
AP DISENGAGE — COUPLED ILS, GLIDESLOPE UNDER 4.5° Verified: Card L-41	Below 200 ft AGL
YAW DAMPER ABOVE 30,000 FT Verified: Card L-42	YD required above FL300; do not override Operation above 30,000 ft MSL is prohibited with the yaw damper disengaged.
RUDDER BIAS — ARMED Verified: PTM App B p772; card ACE-23	Valid N1s, below 2,500 ft AGL, thrust levers in T/O, yaw damper off Confirmed word for word.
RUDDER BIAS — ENGAGES / REMAINS ACTIVE Verified: PTM App B p772	Significant N1 difference; remains until AP quick disconnect, YD engaged, or asymmetry eliminated Confirmed.
EDM — ARMED Verified: PTM App B p772; card ACE-16	Above 30,000 ft MSL and AP engaged
EDM — CANCEL Verified: PTM App B p772; card ACE-18	AP disconnect (trim, AP disc, TCS) AP quick disconnect, manual trim, AP pushbutton on the FD, or TCS — four ways.
TCAS RA/TA INHIBIT Verified: PTM App B p772; card ACE-37 family	Below 1,100 ft AGL climbing, below 900 ft AGL descending Confirmed exactly — radio altitude below 1,100 ft ascending, below 900 ft descending.

Takeoff warning, UMS, water and waste

QUESTION	ANSWER
NO TAKEOFF TRIGGERS Verified: AFM p513, p514	Flaps not 8° or 15°; trim not set; park/emerg brake engaged; accumulator below 2,750 psi; thrust levers at MCT over 2 s; rudder gust lock engaged; airbrake not IN Confirmed against the AFM takeoff configuration list, including the 2,750 psi accumulator threshold and the MCT-for-2-seconds condition.